

Lecture 14: Maps

Heather Mattie, PhD

October 16, 2025

Recipe of the Day!

Butternut Squash Pasta with Walnuts and Sage



~ when you realize tomorrow
is Friday~

Announcements

- Homework #3 due Friday, October 24 by 11:59pm
- Lab will be extended office hours this week
- Mid-course evaluation is still live!

Coding question of the day

Using the `murders` dataset, create a plot of murder rates with 4 boxplots next to each other - one boxplot for each region.

Order the boxplots by median rate (you will need the `reorder` function for this).

Make sure the boxplots are different colors, the axes are informative, and you add a title.

Remove the legend by adding this layer to your plot:
`guides(fill = "none")`

Be sure to run this code first

```
library(dslabs)
library(dplyr)
library(ggplot2)
data("murders")
```

Maps

Maps Roadmap

1. The **maps** package
2. Plotting global life expectancy
3. Plotting US state murder rates
4. Case study: COVID-19 cases

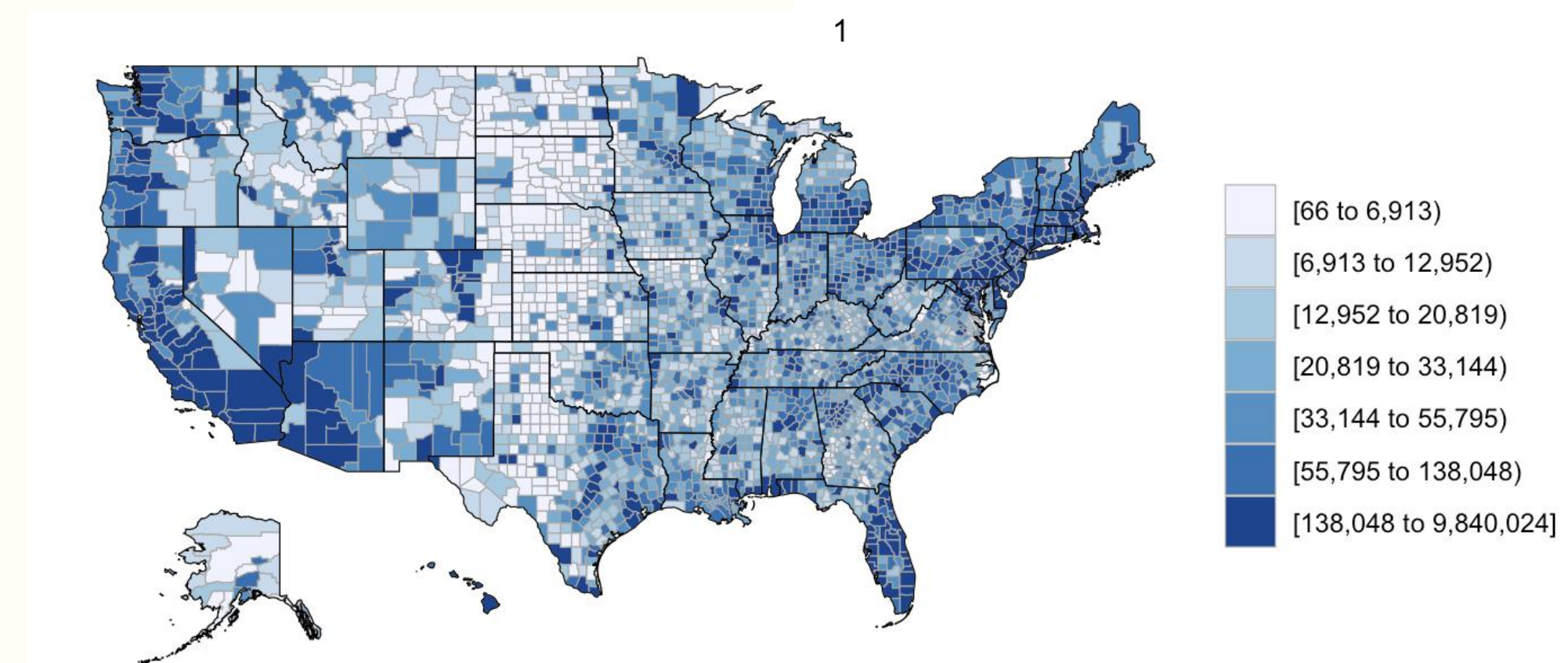
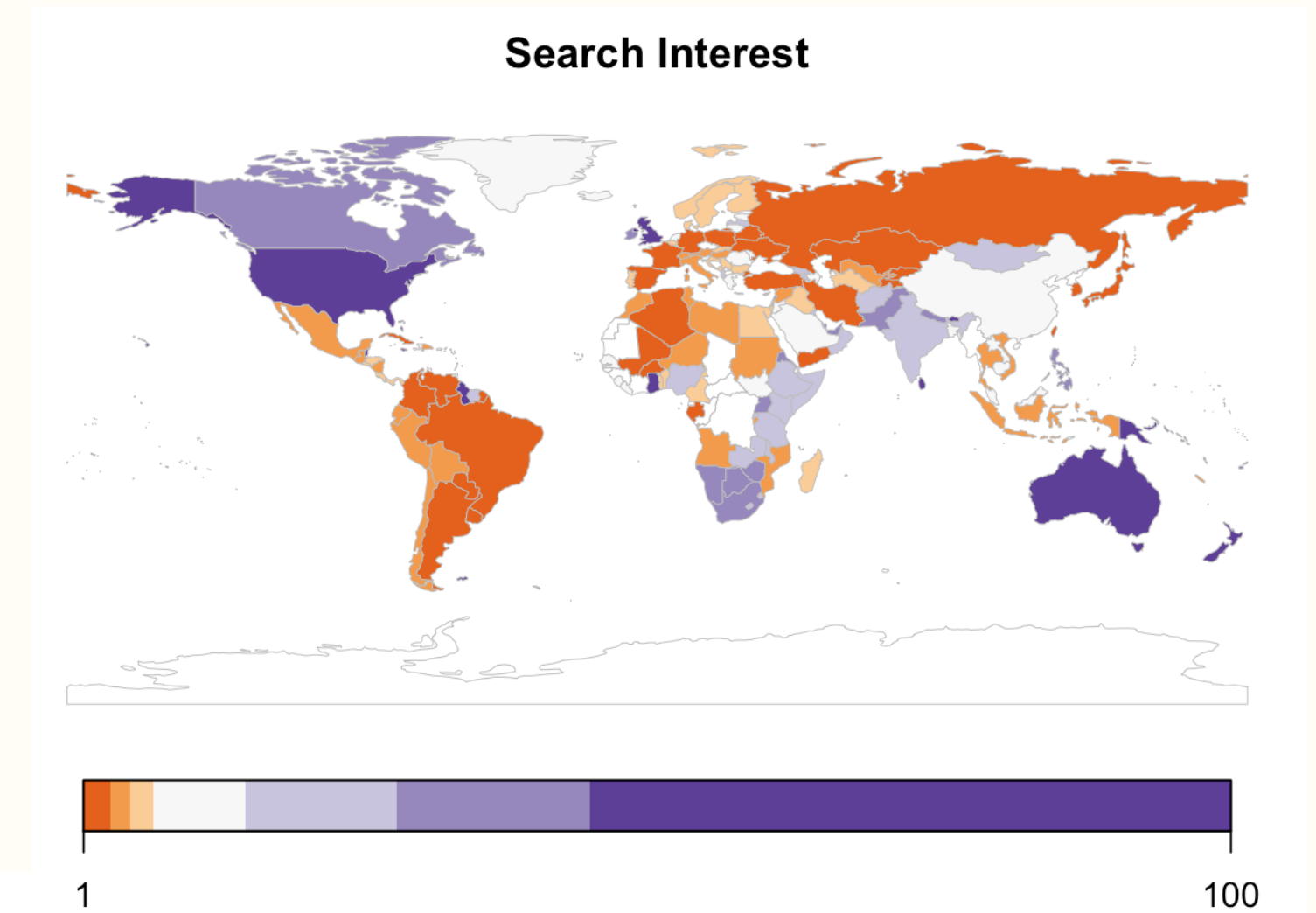
Maps

The **maps** package

Heather Mattie, PhD

Maps

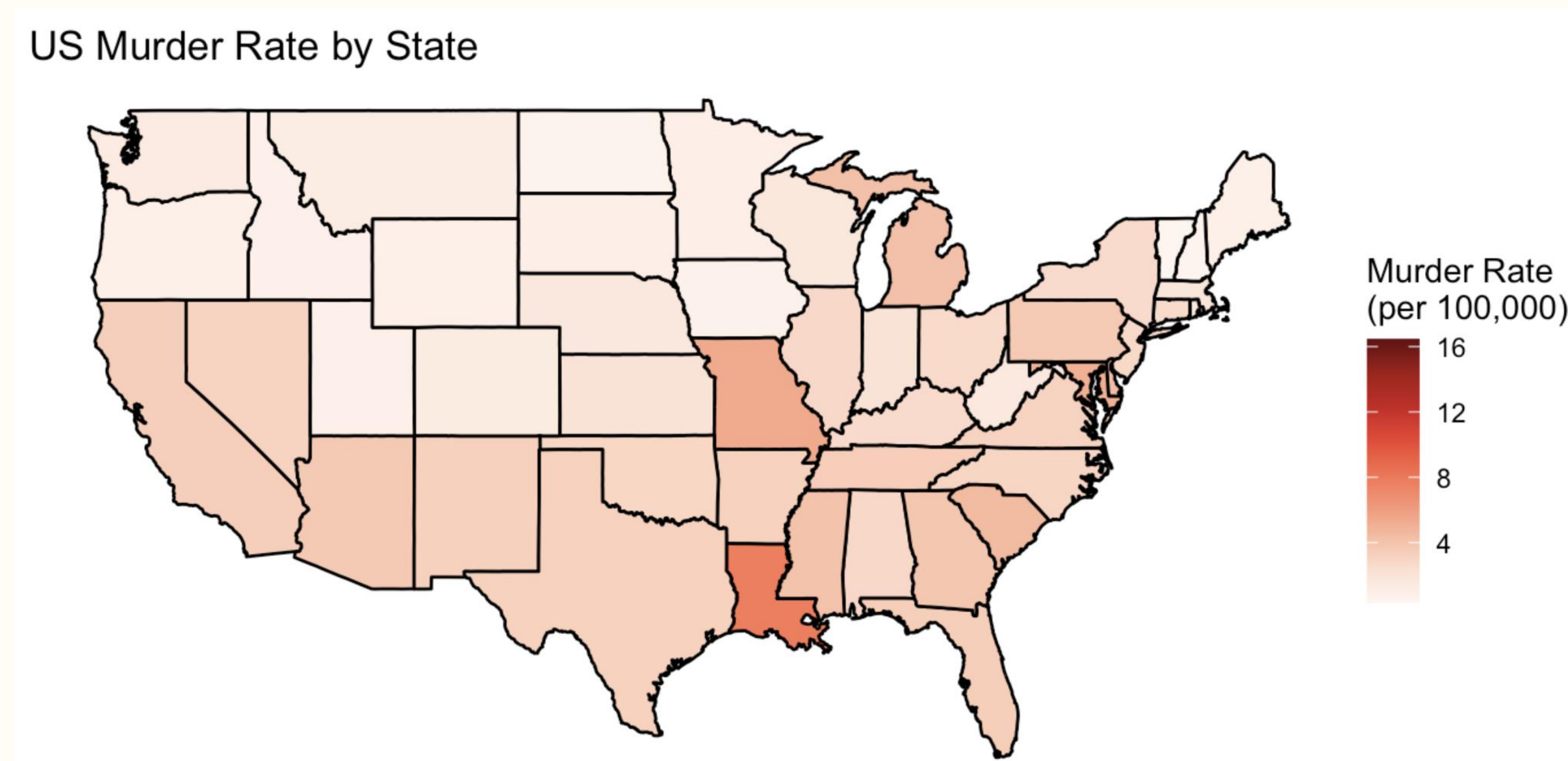
- Plots of **maps** can be very powerful, informative and aesthetically pleasing
- They can also be misleading if not created correctly
- We will use the **maps** package to create plots of maps
- Other packages: **ggspatial**, **sf**, **urbnmapr**, **rnaturalearth**, etc.



maps

The maps package contains outlines of continents, countries, states, and counties / jurisdictions / provinces

- Specify which map to use with `map_data()`
- Loaded as a data frame and can be used with the pipe (`%>%`) and `ggplot`



map_data("world")

- **long** = longitude
- **lat** = latitude
- **group** indicates which countries can be drawn together
- **order** indicates which order to connect the **lat** and **long** dots
- **region** = country
- **subregion** = county, political jurisdictions, etc.

```
library(tidyverse)
library(maps)
world_map <- map_data("world")
head(world_map)
```

##		long	lat	group	order	region	subregion
##	1	-69.89912	12.45200	1	1	Aruba	<NA>
##	2	-69.89571	12.42300	1	2	Aruba	<NA>
##	3	-69.94219	12.43853	1	3	Aruba	<NA>
##	4	-70.00415	12.50049	1	4	Aruba	<NA>
##	5	-70.06612	12.54697	1	5	Aruba	<NA>
##	6	-70.05088	12.59707	1	6	Aruba	<NA>

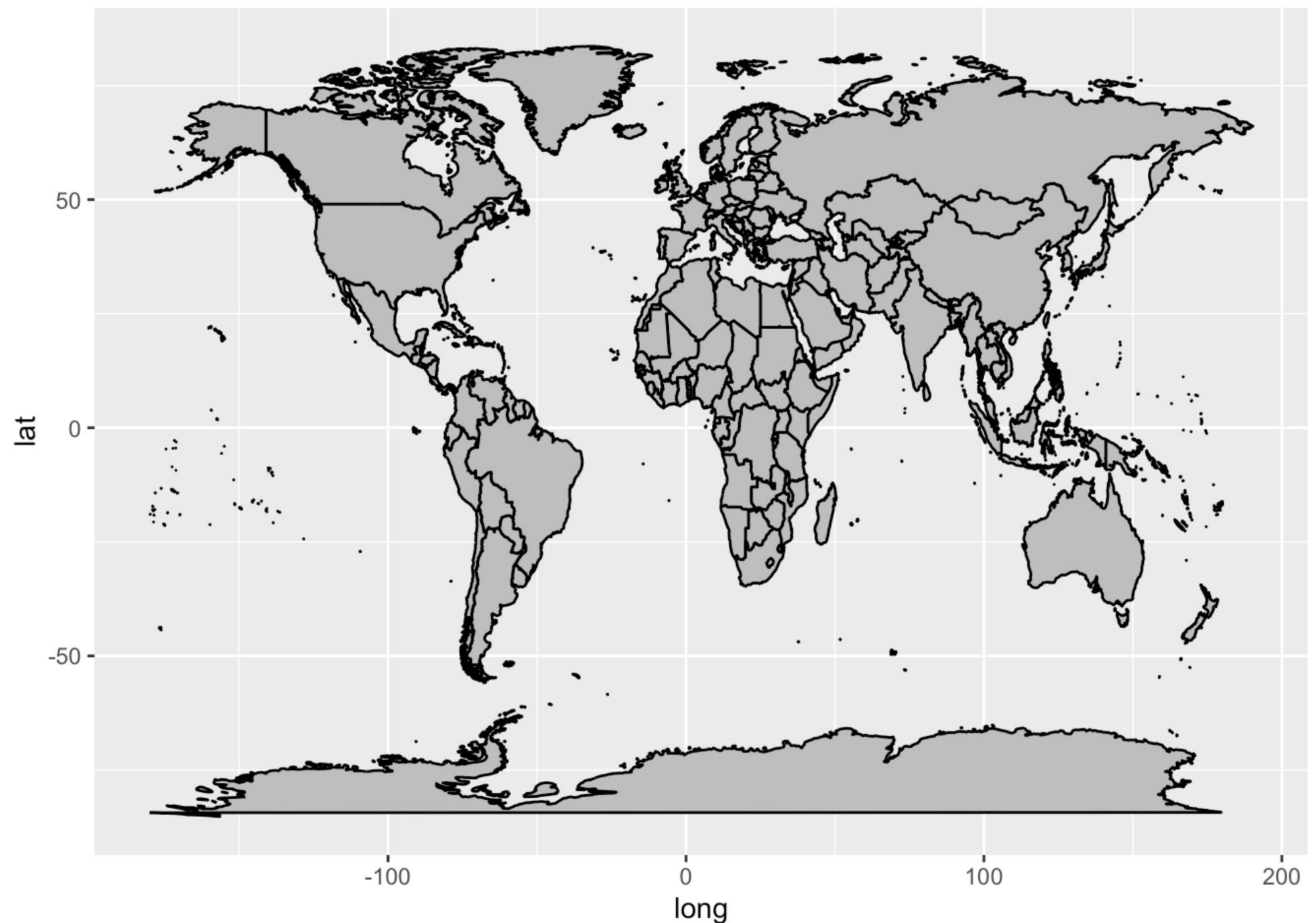
```
world_map %>%
  ggplot(aes(x = long, y = lat, group = group)) +
  geom_polygon(fill = "gray", color = "black")
```


map_data("world")

```
library(tidyverse)
library(maps)
world_map <- map_data("world")
head(world_map)
```

```
##           long      lat group order region subregion
## 1 -69.89912 12.45200     1     1  Aruba    <NA>
## 2 -69.89571 12.42300     1     2  Aruba    <NA>
## 3 -69.94219 12.43853     1     3  Aruba    <NA>
## 4 -70.00415 12.50049     1     4  Aruba    <NA>
## 5 -70.06612 12.54697     1     5  Aruba    <NA>
## 6 -70.05088 12.59707     1     6  Aruba    <NA>
```

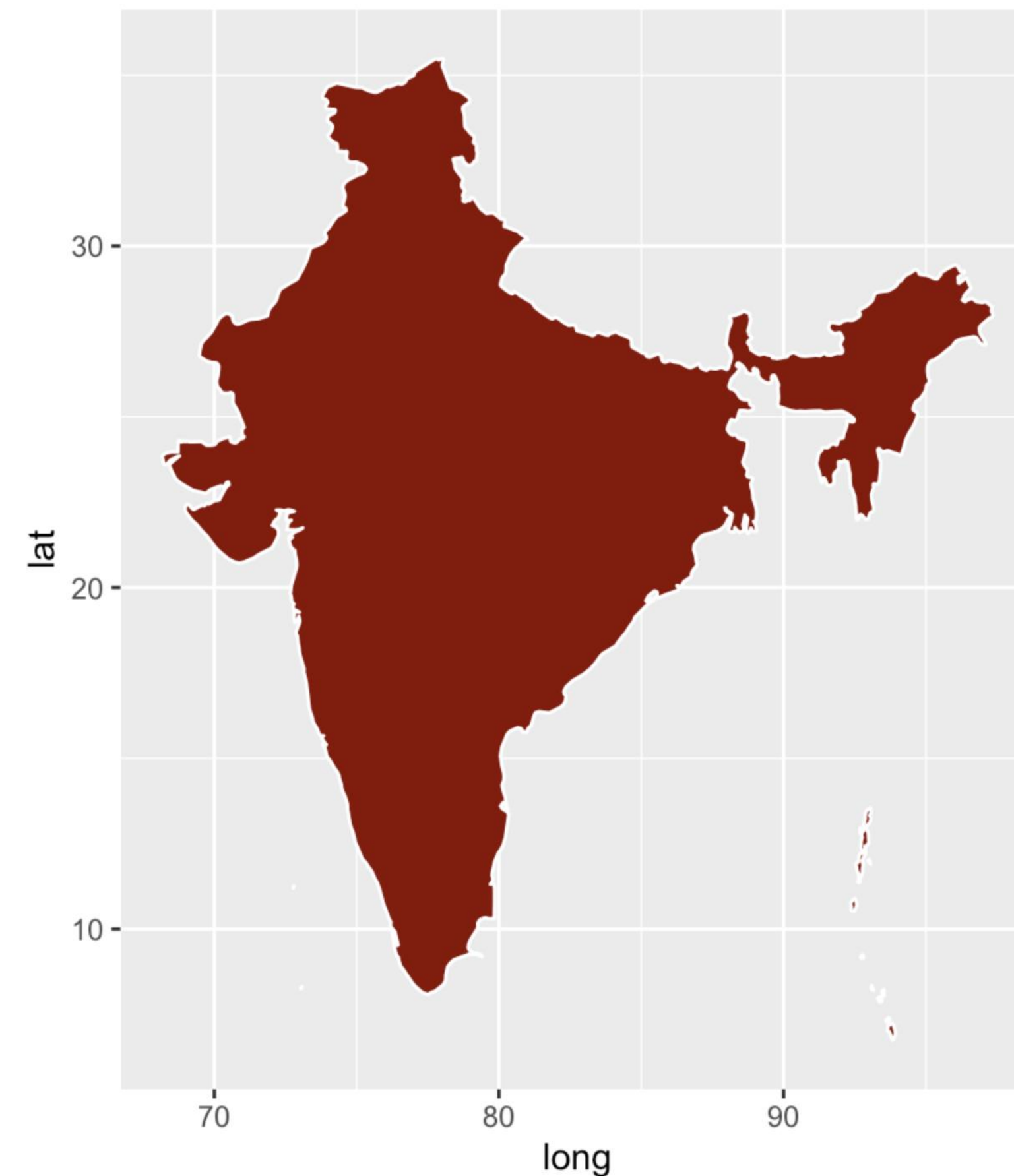
```
world_map %>%
  ggplot(aes(x = long, y = lat, group = group)) +
  geom_polygon(fill = "gray", color = "black")
```



Countries

```
india_map <- map_data("world", "India")

india_map %>%
  ggplot(aes(x = long, y = lat, group = group)) +
  geom_polygon(fill = "darkred", color = "white") +
  coord_fixed(1.2)
```

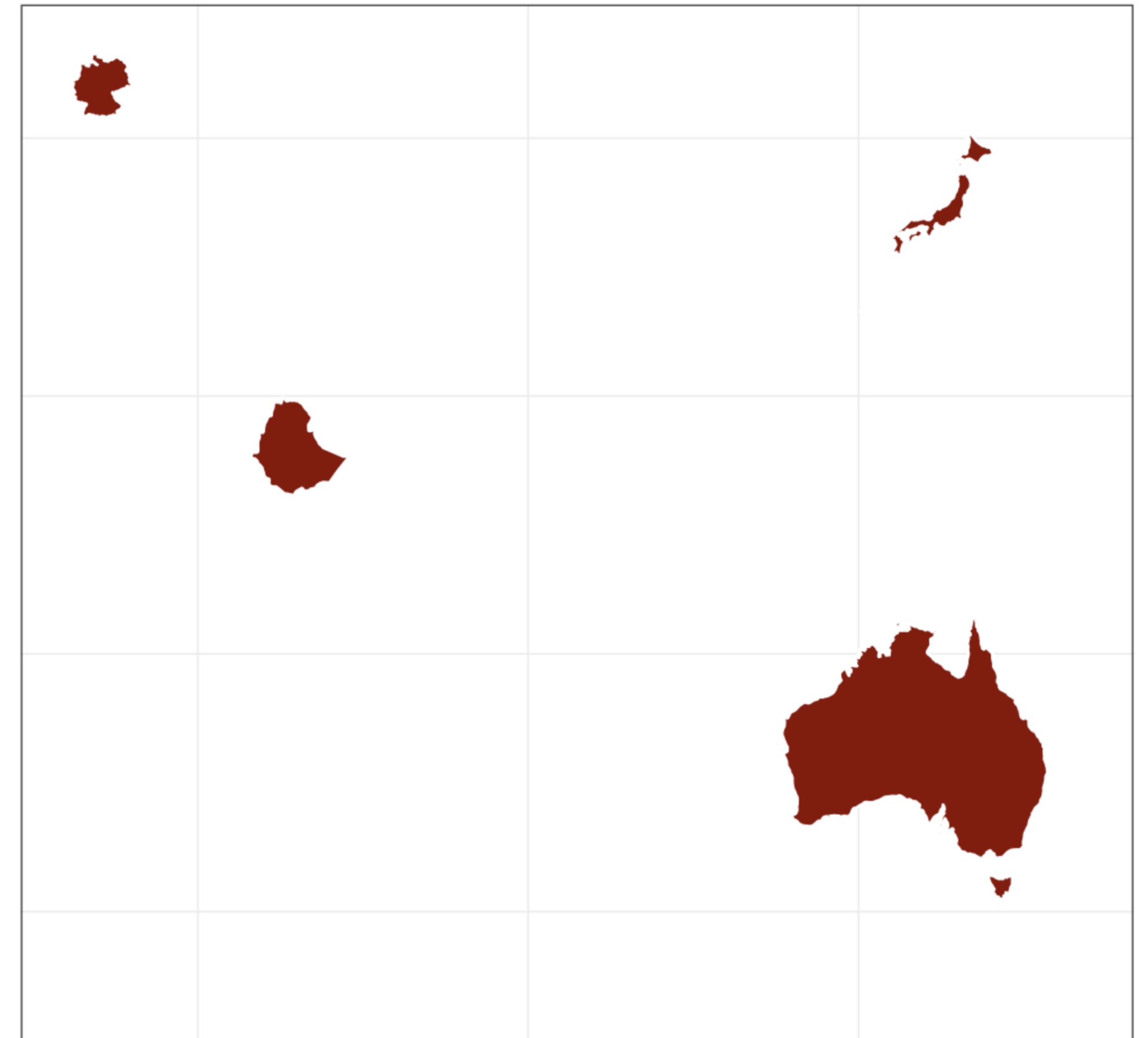


Countries

```
theme_set(theme_bw())

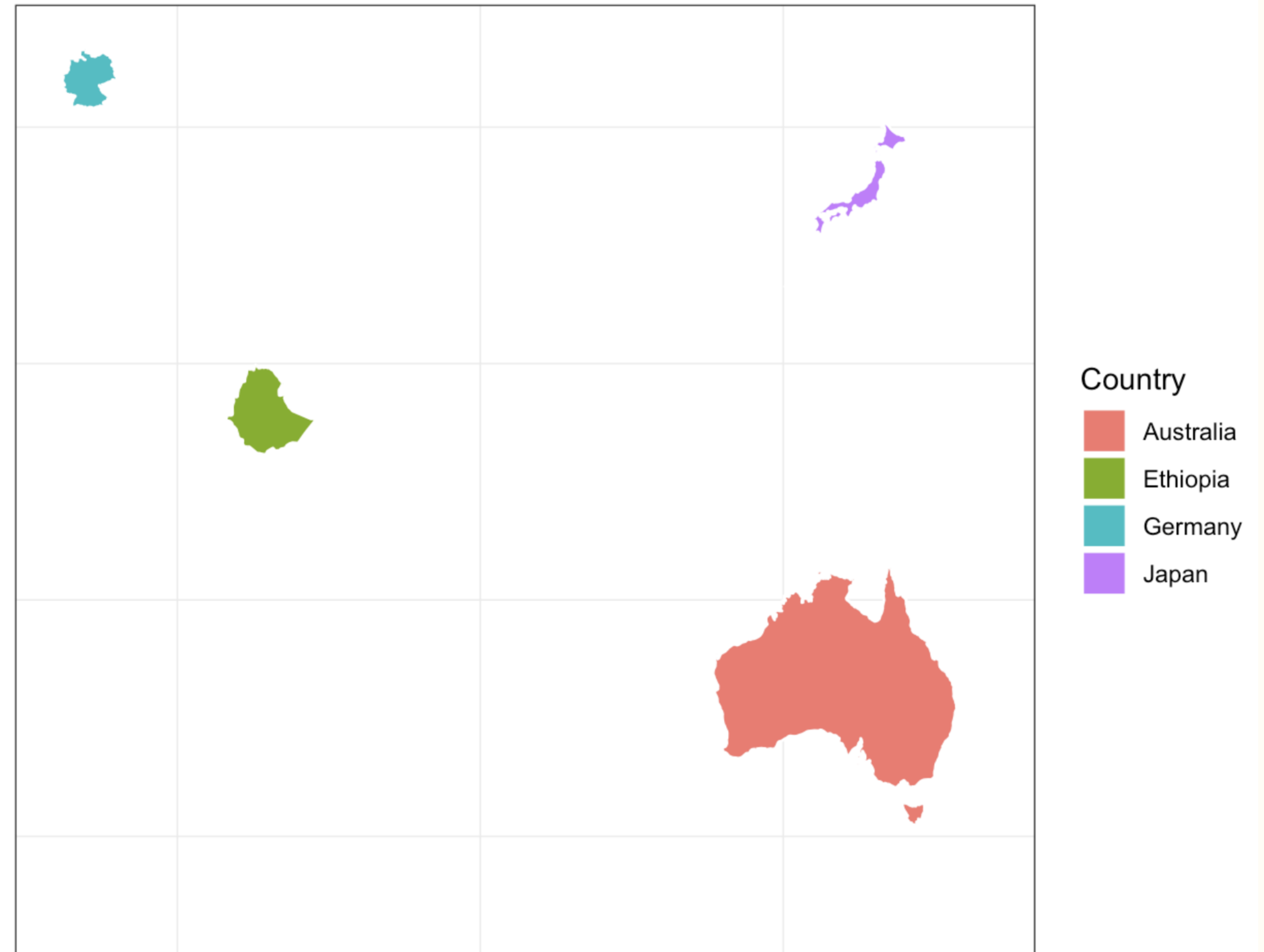
multiple_countries_map <- map_data("world",
                                   c("Japan", "Ethiopia", "Australia", "Germany"))

multiple_countries_map %>%
  ggplot(aes(x = long, y = lat, group = group)) +
  geom_polygon(fill = "darkred", color = "white") +
  theme(panel.grid.major = element_blank(),
        panel.background = element_blank(),
        axis.title = element_blank(),
        axis.text = element_blank(),
        axis.ticks = element_blank()) +
  coord_fixed(1.3)
```



Countries

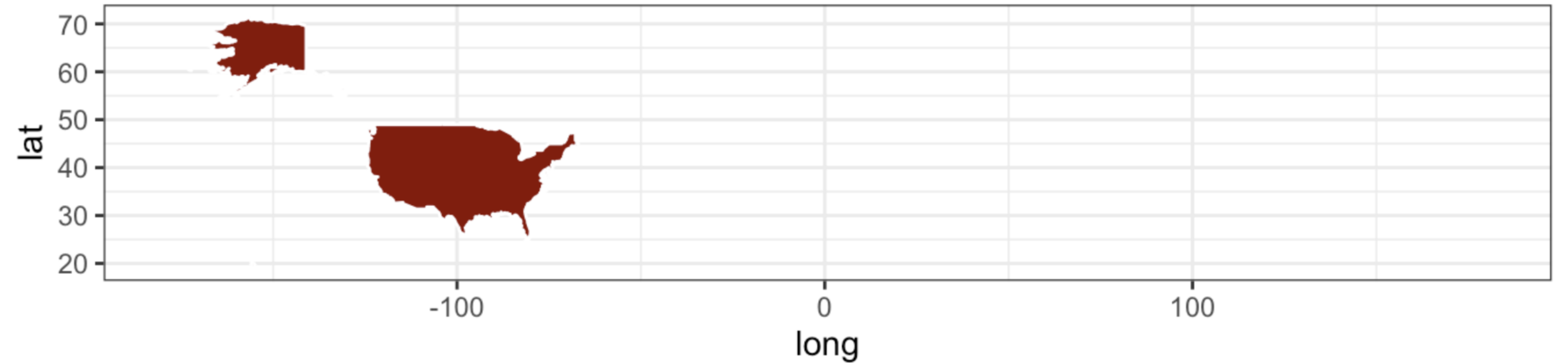
```
multiple_countries_map %>%  
  ggplot(aes(x = long, y = lat, group = group, fill = region)) +  
  geom_polygon(color = "white") +  
  theme(panel.grid.major = element_blank(),  
        panel.background = element_blank(),  
        axis.title = element_blank(),  
        axis.text = element_blank(),  
        axis.ticks = element_blank()) +  
  labs(fill = "Country") +  
  coord_fixed(1.3)
```



US maps

```
us_map <- map_data("world", "usa")

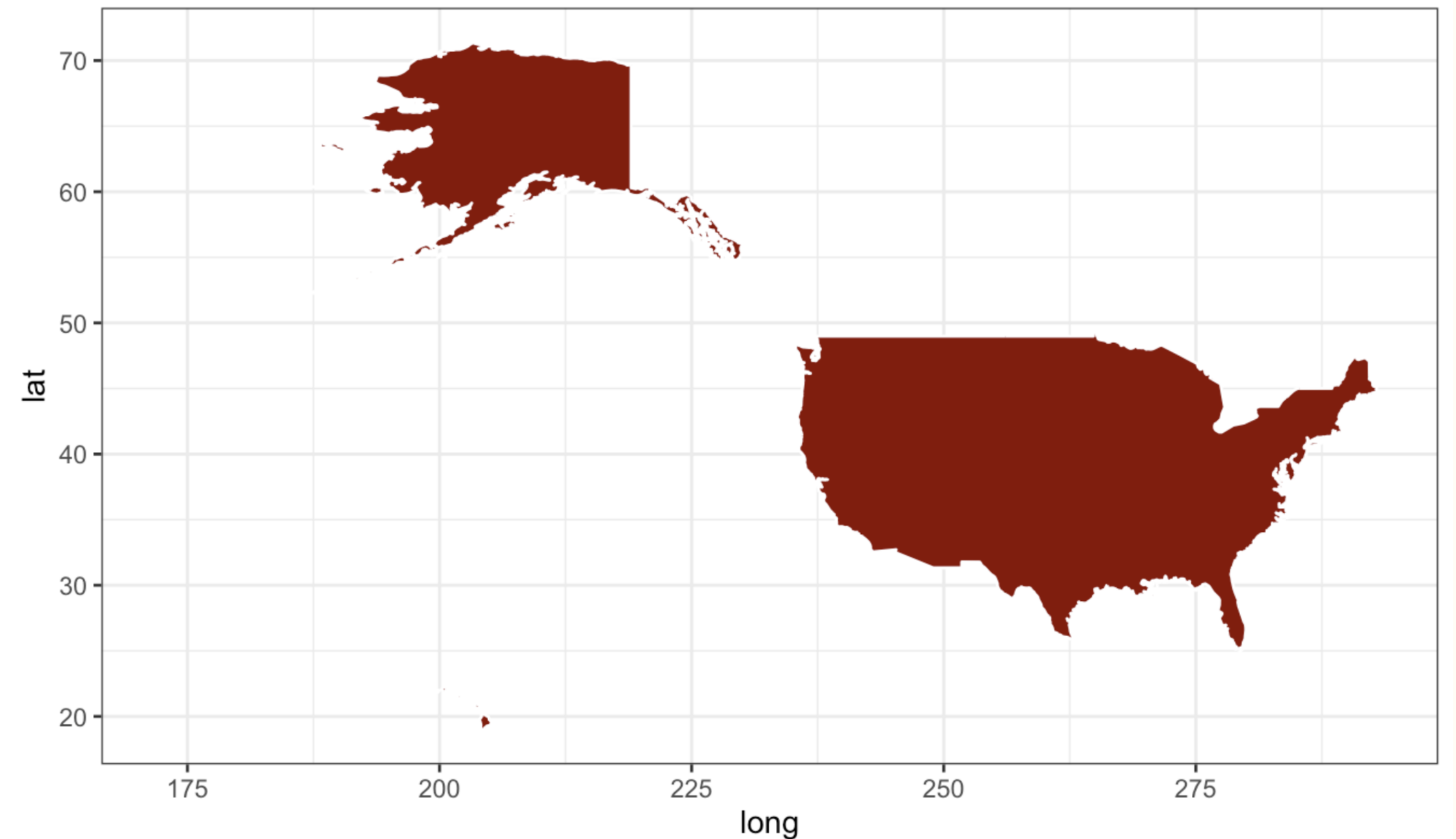
us_map %>%
  ggplot(aes(x = long, y = lat, group = group)) +
  geom_polygon(fill = "darkred", color = "white") +
  coord_fixed(1.3)
```



US maps

```
better_us_map <- map_data("world2", "USA")

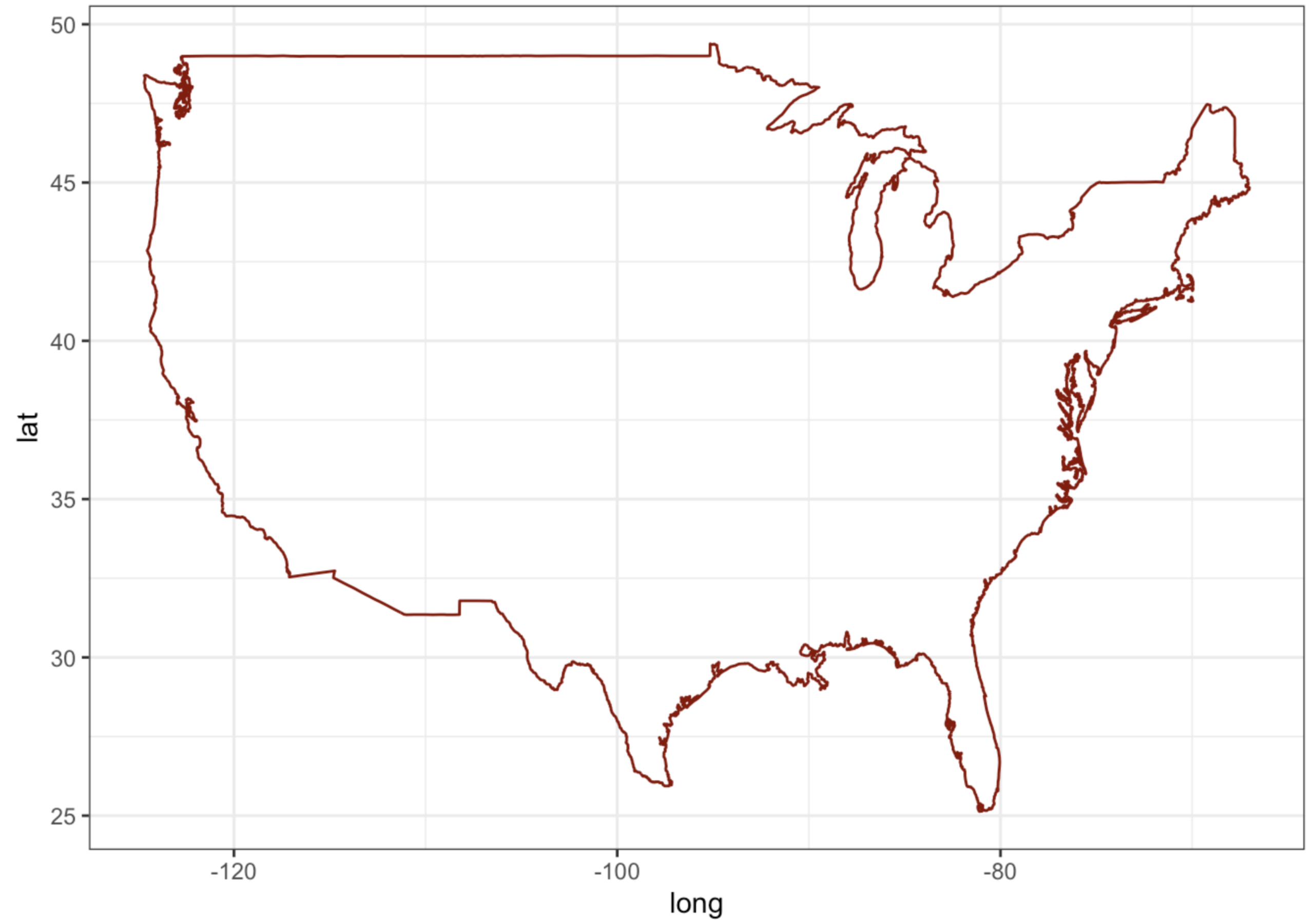
better_us_map %>%
  ggplot(aes(x = long, y = lat, group = group)) +
  geom_polygon(fill = "darkred", color = "white") +
  coord_fixed(1.3)
```



US maps

```
usa <- map_data("usa")

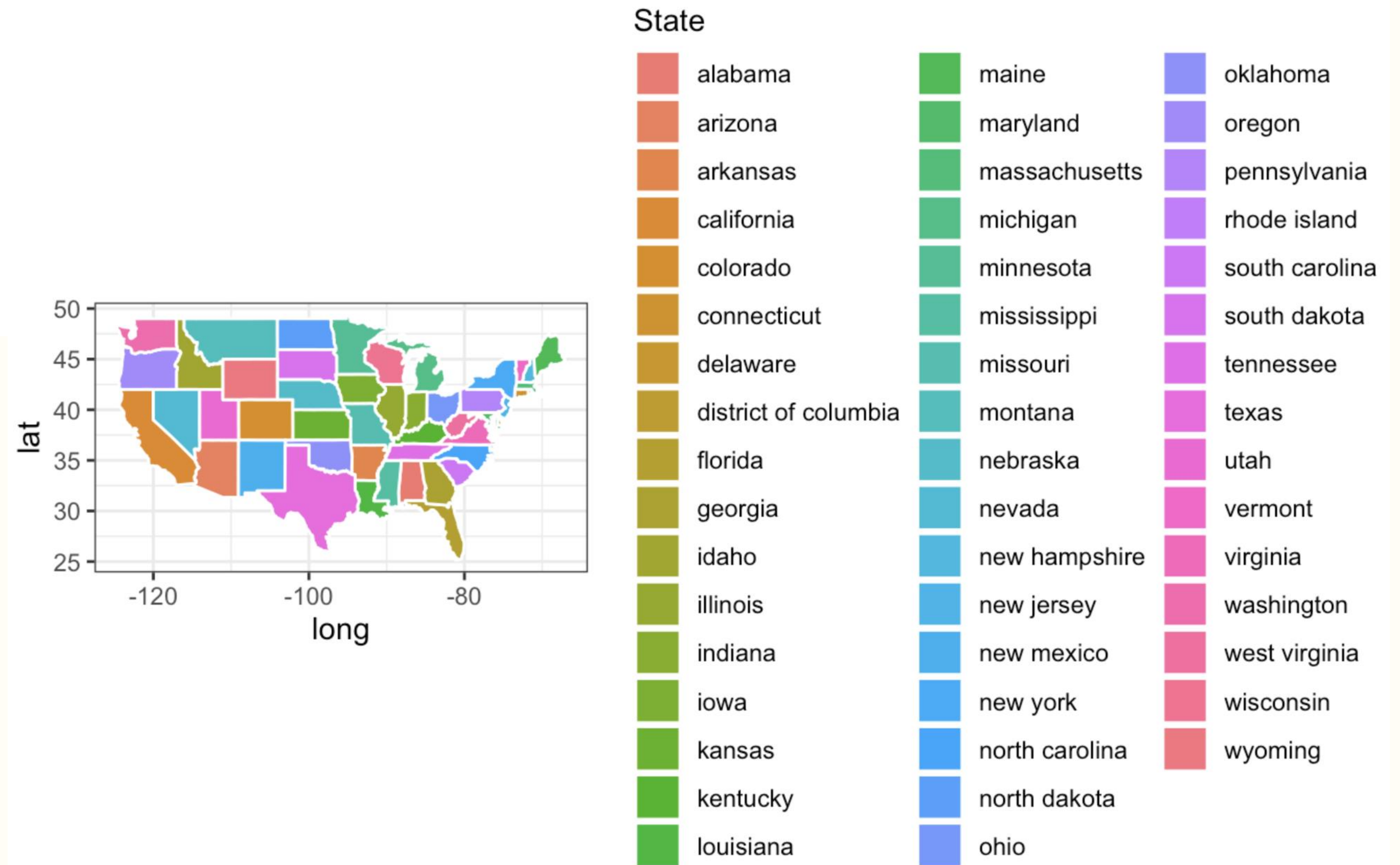
usa %>%
  ggplot(aes(x = long, y = lat, group = group)) +
  geom_polygon(color = "darkred", fill = NA)
```



US states

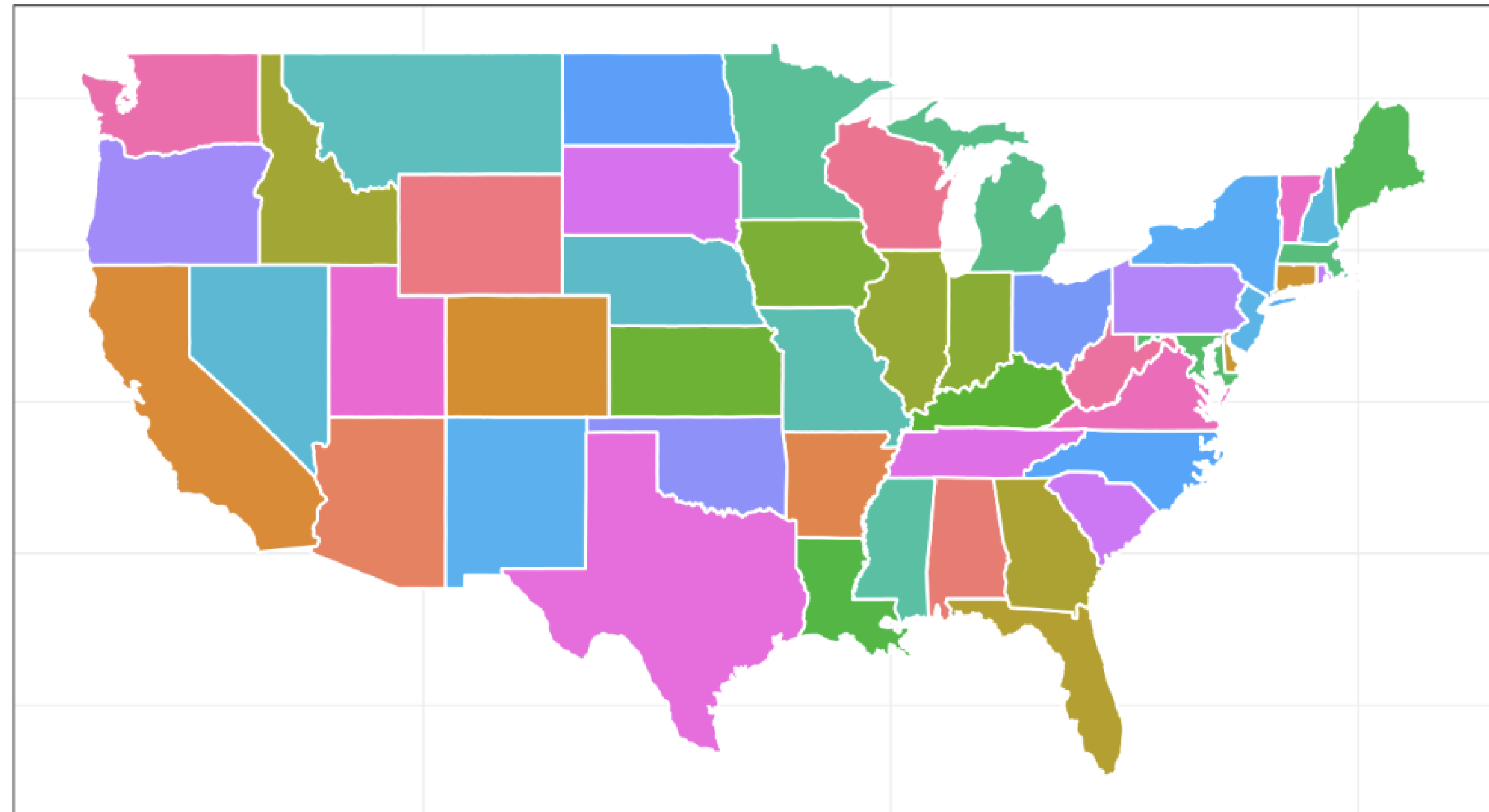
```
us_states <- map_data("state")

us_states %>%
  ggplot(aes(x = long, y = lat, fill = region, group = group)) +
  geom_polygon(color = "white") +
  labs(fill = "State") +
  coord_fixed(1.3)
```



US states

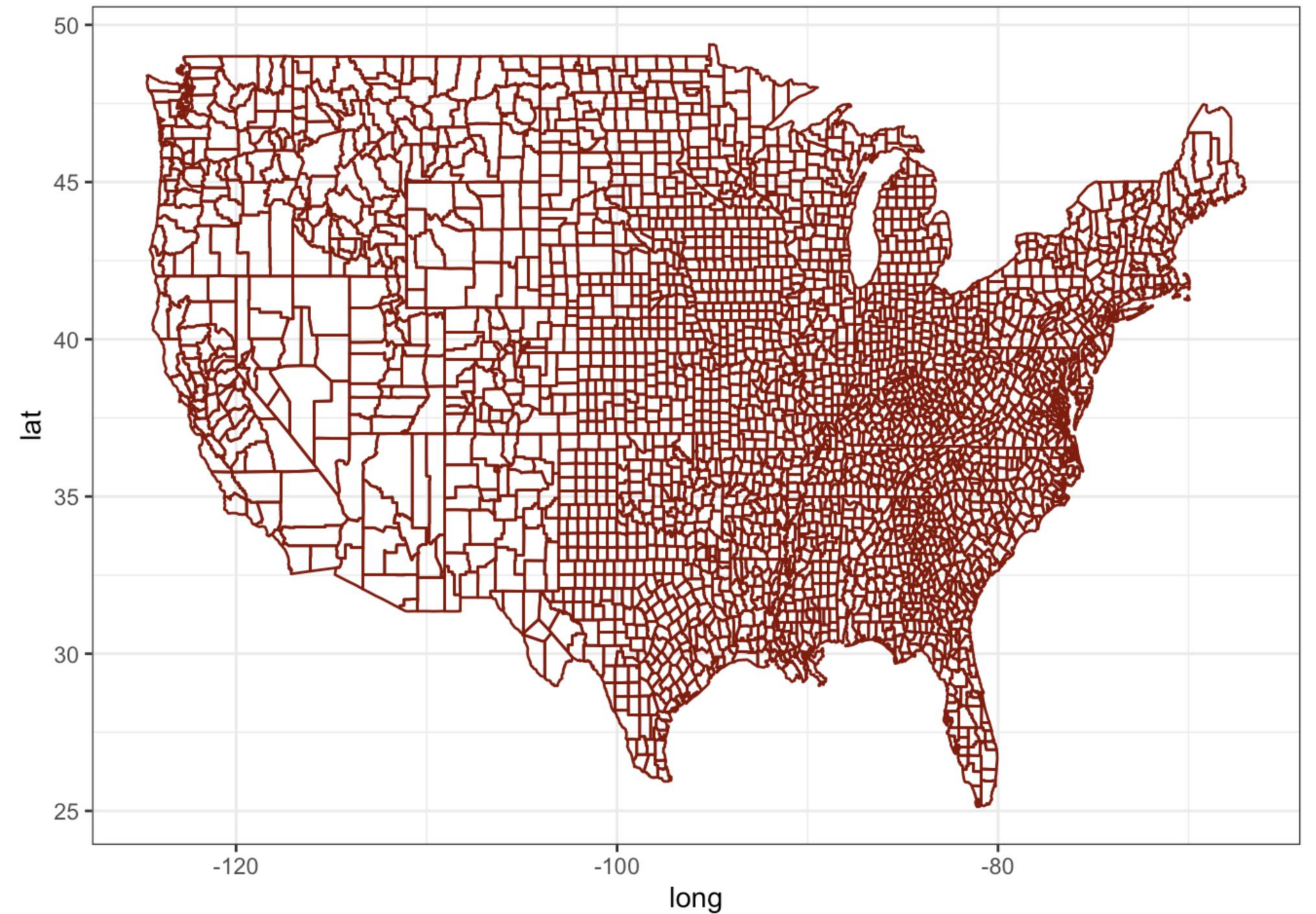
```
us_states %>%  
  ggplot(aes(x = long, y = lat, fill = region, group = group)) +  
  geom_polygon(color = "white") +  
  theme(panel.grid.major = element_blank(),  
        panel.background = element_blank(),  
        axis.title = element_blank(),  
        axis.text = element_blank(),  
        axis.ticks = element_blank()) +  
  coord_fixed(1.3) +  
  guides(fill = "none")
```



US counties

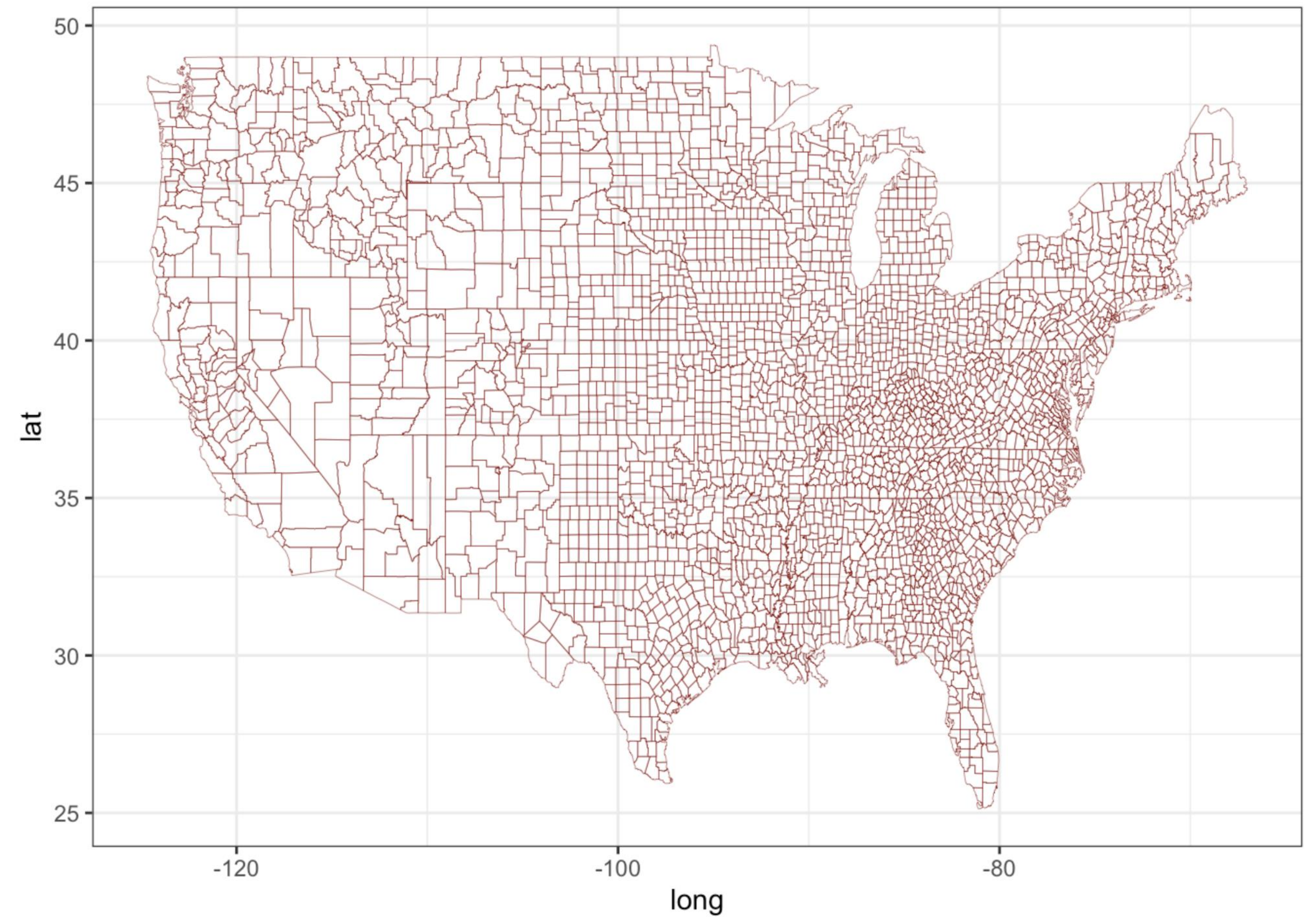
```
AllCounty <- map_data("county")
```

```
AllCounty %>%  
  ggplot(aes(x = long, y = lat, group = group)) +  
  geom_polygon(color = "darkred", fill = NA)
```



US counties

```
AllCounty %>%  
  ggplot(aes(x = long, y = lat, group = group)) +  
  geom_polygon(color = "darkred", fill = NA, linewidth = .1)
```



Summary

- The **maps** package can be used to produce maps of continents, countries, states, counties, and provinces
- The **map_data()** function loads map data
- Color or labels should be used instead of a legend if there are too many categories (countries, states, etc.) represented in the map

Maps

Plotting global life expectancy

Heather Mattie, PhD

Plotting global life expectancy

Task: plot a map displaying the life expectancy in 2015 for countries with data in the **gapminder** dataset

```
library(plyr)
library(tidyverse)
library(maps)
library(dslabs)

data(gapminder)
gapminder_2015 <- gapminder %>%
  filter(year == 2015)

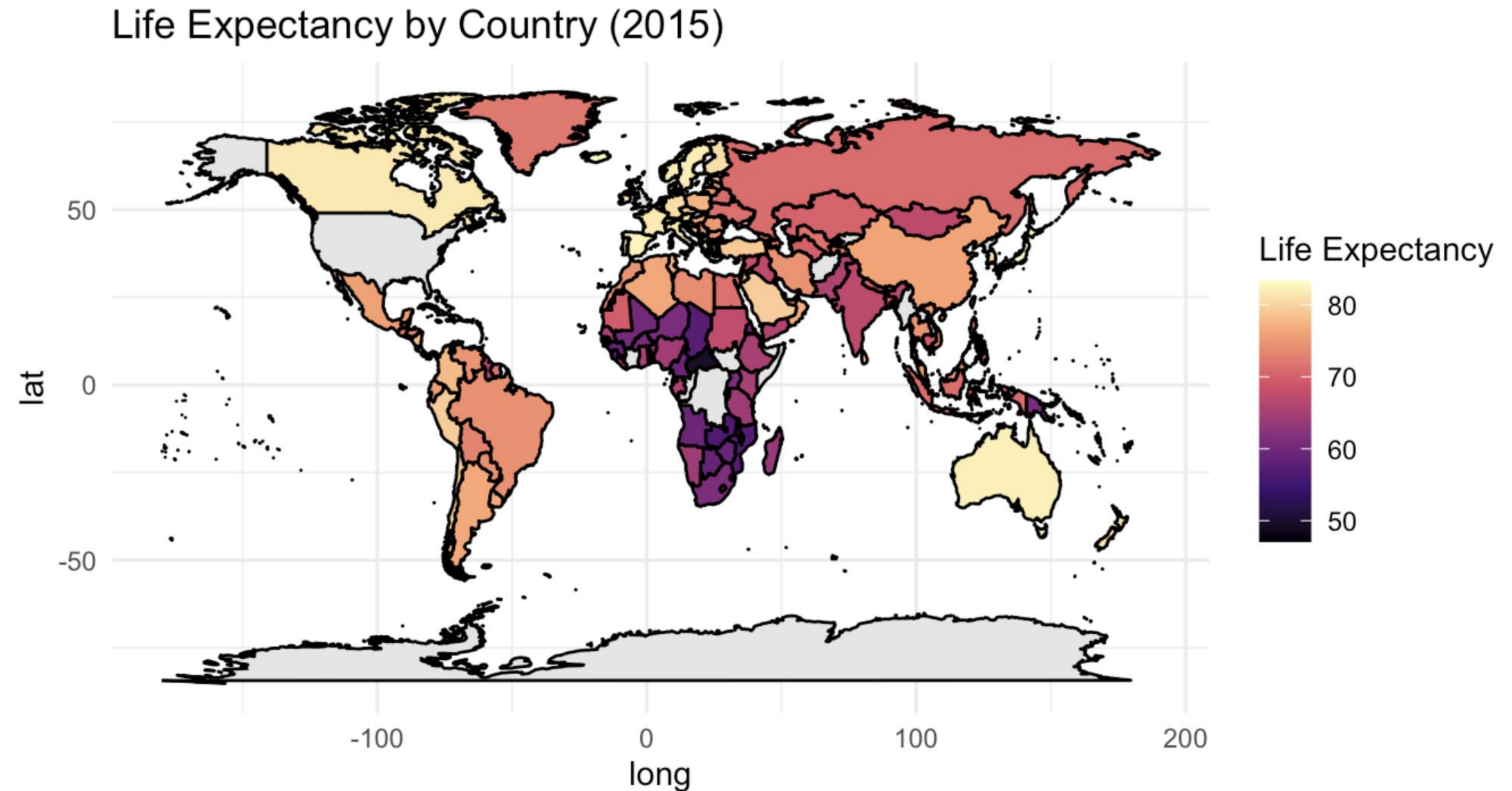
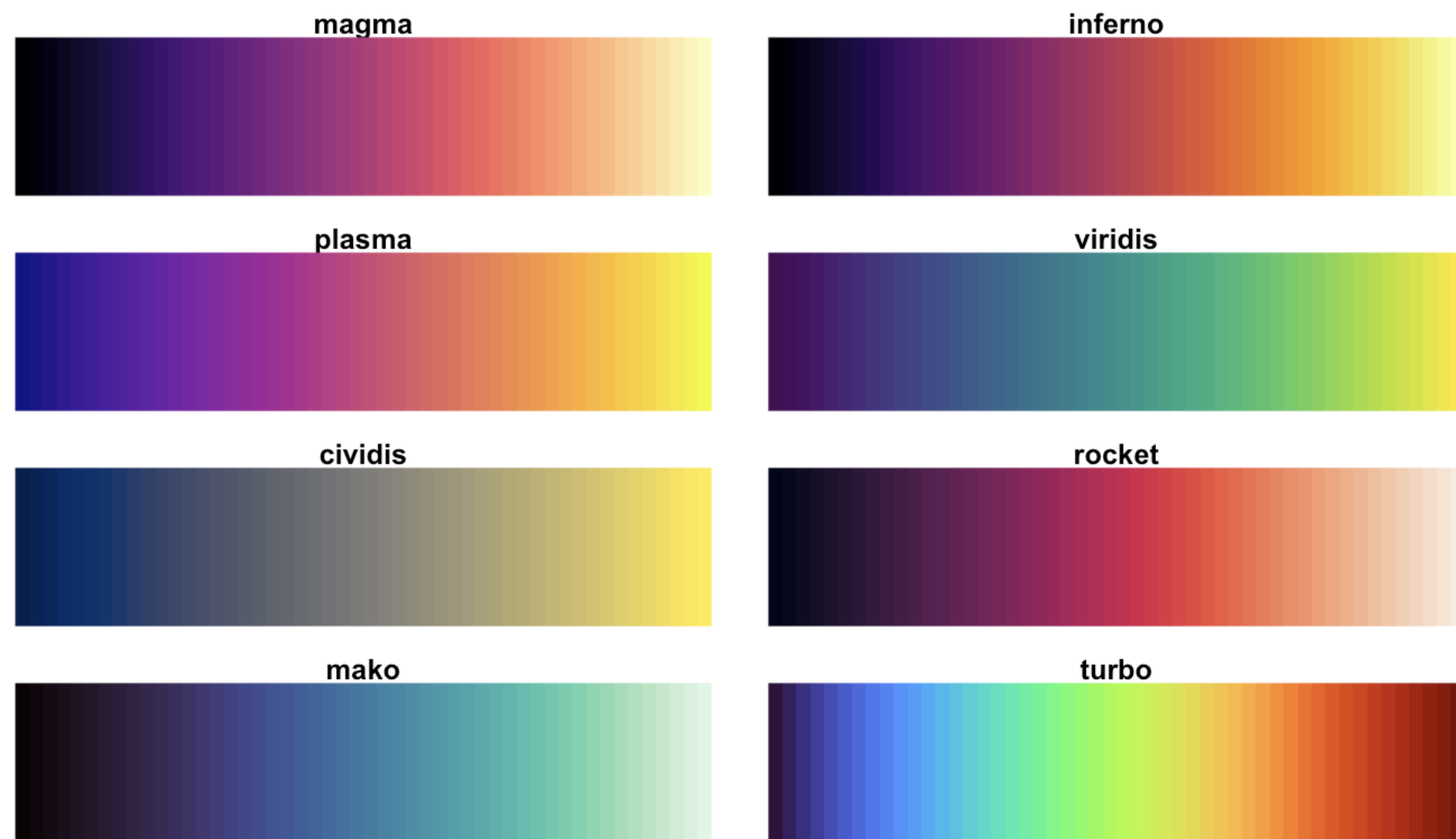
world_map <- map_data("world")
merged_world <- left_join(world_map, gapminder_2015,
                          by = c("region" = "country"))

head(merged_world)
```

```
##      long      lat group order region subregion year infant_mortality
## 1 -69.89912 12.45200     1     1  Aruba      <NA> 2015              NA
## 2 -69.89571 12.42300     1     2  Aruba      <NA> 2015              NA
## 3 -69.94219 12.43853     1     3  Aruba      <NA> 2015              NA
## 4 -70.00415 12.50049     1     4  Aruba      <NA> 2015              NA
## 5 -70.06612 12.54697     1     5  Aruba      <NA> 2015              NA
## 6 -70.05088 12.59707     1     6  Aruba      <NA> 2015              NA
##      life_expectancy fertility population gdp continent region.y
## 1             75.72      1.66     103889   NA  Americas Caribbean
## 2             75.72      1.66     103889   NA  Americas Caribbean
## 3             75.72      1.66     103889   NA  Americas Caribbean
## 4             75.72      1.66     103889   NA  Americas Caribbean
## 5             75.72      1.66     103889   NA  Americas Caribbean
## 6             75.72      1.66     103889   NA  Americas Caribbean
```

Plotting global life expectancy

```
ggplot(merged_world, aes(long, lat, group = group,  
                          fill = life_expectancy)) +  
  geom_polygon(color = "black") +  
  scale_fill_viridis_c(option = "magma",  
                       na.value = "gray90") +  
  theme_minimal() +  
  labs(title = "Life Expectancy by Country (2015)",  
       fill = "Life Expectancy") +  
  coord_fixed(1.3)
```



Data inconsistencies

```
issues <- anti_join(world_map, gapminder_2015, by = c("region" = "country"))
levels(as.factor(issues$region))
```

```
## [1] "Afghanistan"      "American Samoa"
## [3] "Andorra"          "Anguilla"
## [5] "Antarctica"       "Antigua"
## [7] "Ascension Island" "Azores"
## [9] "Barbuda"          "Bermuda"
## [11] "Bonaire"          "Canary Islands"
## [13] "Cayman Islands"   "Chagos Archipelago"
## [15] "Christmas Island" "Cocos Islands"
## [17] "Cook Islands"     "Curacao"
## [19] "Democratic Republic of the Congo" "Dominica"
## [21] "Falkland Islands" "Faroe Islands"
## [23] "French Guiana"    "French Southern and Antarctic Lands"
## [25] "Grenadines"       "Guadeloupe"
## [27] "Guam"             "Guernsey"
## [29] "Heard Island"     "Isle of Man"
## [31] "Ivory Coast"      "Jersey"
## [33] "Kosovo"           "Kyrgyzstan"
## [35] "Laos"             "Liechtenstein"
## [37] "Madeira Islands"  "Marshall Islands"
## [39] "Martinique"       "Mayotte"
## [41] "Micronesia"       "Monaco"
## [43] "Montserrat"       "Myanmar"
## [45] "Nauru"            "Nevis"
```

```
levels(gapminder_2015$country)
```

```
## [1] "Albania"          "Algeria"
## [3] "Angola"           "Antigua and Barbuda"
## [5] "Argentina"        "Armenia"
## [7] "Aruba"            "Australia"
## [9] "Austria"          "Azerbaijan"
## [11] "Bahamas"          "Bahrain"
## [13] "Bangladesh"       "Barbados"
## [15] "Belarus"          "Belgium"
## [17] "Belize"           "Benin"
## [19] "Bhutan"           "Bolivia"
## [21] "Bosnia and Herzegovina" "Botswana"
## [23] "Brazil"           "Brunei"
## [25] "Bulgaria"         "Burkina Faso"
## [27] "Burundi"          "Cambodia"
## [29] "Cameroon"         "Canada"
## [31] "Cape Verde"       "Central African Republic"
## [33] "Chad"             "Chile"
## [35] "China"            "Colombia"
## [37] "Comoros"          "Congo, Dem. Rep."
## [39] "Congo, Rep."      "Costa Rica"
## [41] "Cote d'Ivoire"    "Croatia"
## [43] "Cuba"             "Cyprus"
## [45] "Czech Republic"  "Denmark"
## [47] "Djibouti"         "Dominican Republic"
## [49] "Ecuador"          "Egypt"
## [51] "El Salvador"      "Equatorial Guinea"
## [53] "Eritrea"          "Estonia"
```



Data inconsistencies

```
issues <- anti_join(world_map, gapminder_2015, by = c("region" = "country"))
levels(as.factor(issues$region))
```

```
## [1] "Afghanistan"      "American Samoa"
## [3] "Andorra"          "Anguilla"
## [5] "Antarctica"       "Antigua"
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## [21] "Falkland Islands" "Faroe Islands"
## [23] "French Guiana"    "French Southern and Antarctic Lands"
## [25] "Grenadines"       "Guadeloupe"
## [27] "Guam"             "Guernsey"
## [29] "Heard Island"     "Isle of Man"
## [31] "Ivory Coast"      "Jersey"
## [33] "Kosovo"           "Kyrgyzstan"
## [35] "Laos"             "Liechtenstein"
## [37] "Madeira Islands"  "Marshall Islands"
## [39] "Martinique"       "Mayotte"
## [41] "Micronesia"       "Monaco"
## [43] "Montserrat"       "Myanmar"
## [45] "Nauru"            "Nevis"
```

```
levels(gapminder_2015$country)
```

```
## [1] "Albania"          "Algeria"
## [3] "Angola"           "Antigua and Barbuda"
## [5] "Argentina"        "Armenia"
## [7] "Aruba"            "Australia"
## [9] "Austria"          "Azerbaijan"
## [11] "Bahamas"          "Bahrain"
## [13] "Bangladesh"       "Barbados"
## [15] "Belarus"          "Belgium"
## [17] "Belize"           "Benin"
## [19] "Bhutan"           "Bolivia"
## [21] "Bosnia and Herzegovina" "Botswana"
## [23] "Brazil"           "Brunei"
## [25] "Bulgaria"         "Burkina Faso"
## [27] "Burundi"          "Cambodia"
## [29] "Cameroon"         "Canada"
## [31] "Cape Verde"       "Central African Republic"
## [33] "Chad"             "Chile"
## [35] "China"            "Colombia"
## [37] "Comoros"          "Congo, Dem. Rep."
## [39] "Congo, Rep."      "Costa Rica"
## [41] "Cote d'Ivoire"    "Croatia"
## [43] "Cuba"             "Cyprus"
## [45] "Czech Republic"  "Denmark"
## [47] "Djibouti"         "Dominican Republic"
## [49] "Ecuador"          "Egypt"
## [51] "El Salvador"      "Equatorial Guinea"
## [53] "Eritrea"          "Estonia"
```



Data inconsistencies

```
issues <- anti_join(world_map, gapminder_2015, by = c("region" = "country"))
levels(as.factor(issues$region))
```

```
## [1] "Afghanistan"      "American Samoa"
## [3] "Andorra"          "Anguilla"
## [5] "Antarctica"       "Antigua"
## [7] "Ascension Island" "Azores"
## [9] "Barbuda"          "Bermuda"
## [11] "Bonaire"          "Canary Islands"
## [13] "Cayman Islands"   "Chagos Archipelago"
## [15] "Christmas Island" "Cocos Islands"
## [17] "Cook Islands"     "Curacao"
## [19] "Democratic Republic of the Congo" "Dominica"
## [21] "Falkland Islands" "Faroe Islands"
## [23] "French Guiana"    "French Southern and Antarctic Lands"
## [25] "Grenadines"       "Guadeloupe"
## [27] "Guam"             "Guernsey"
## [29] "Heard Island"     "Isle of Man"
## [31] "Ivory Coast"       "Jersey"
## [33] "Kosovo"            "Kyrgyzstan"
## [35] "Laos"              "Liechtenstein"
## [37] "Madeira Islands"  "Marshall Islands"
## [39] "Martinique"       "Mayotte"
## [41] "Micronesia"       "Monaco"
## [43] "Montserrat"       "Myanmar"
## [45] "Nauru"             "Nevis"
```

```
levels(gapminder_2015$country)
```

```
## [1] "Albania"          "Algeria"
## [3] "Angola"           "Antigua and Barbuda"
## [5] "Argentina"        "Armenia"
## [7] "Aruba"            "Australia"
## [9] "Austria"          "Azerbaijan"
## [11] "Bahamas"          "Bahrain"
## [13] "Bangladesh"       "Barbados"
## [15] "Belarus"          "Belgium"
## [17] "Belize"           "Benin"
## [19] "Bhutan"           "Bolivia"
## [21] "Bosnia and Herzegovina" "Botswana"
## [23] "Brazil"           "Brunei"
## [25] "Bulgaria"         "Burkina Faso"
## [27] "Burundi"          "Cambodia"
## [29] "Cameroon"         "Canada"
## [31] "Cape Verde"       "Central African Republic"
## [33] "Chad"             "Chile"
## [35] "China"            "Colombia"
## [37] "Comoros"          "Congo, Dem. Rep."
## [39] "Congo, Rep."      "Costa Rica"
## [41] "Cote d'Ivoire"    "Croatia"
## [43] "Cuba"             "Cyprus"
## [45] "Czech Republic"  "Denmark"
## [47] "Djibouti"         "Dominican Republic"
## [49] "Ecuador"          "Egypt"
## [51] "El Salvador"      "Equatorial Guinea"
## [53] "Eritrea"          "Estonia"
```

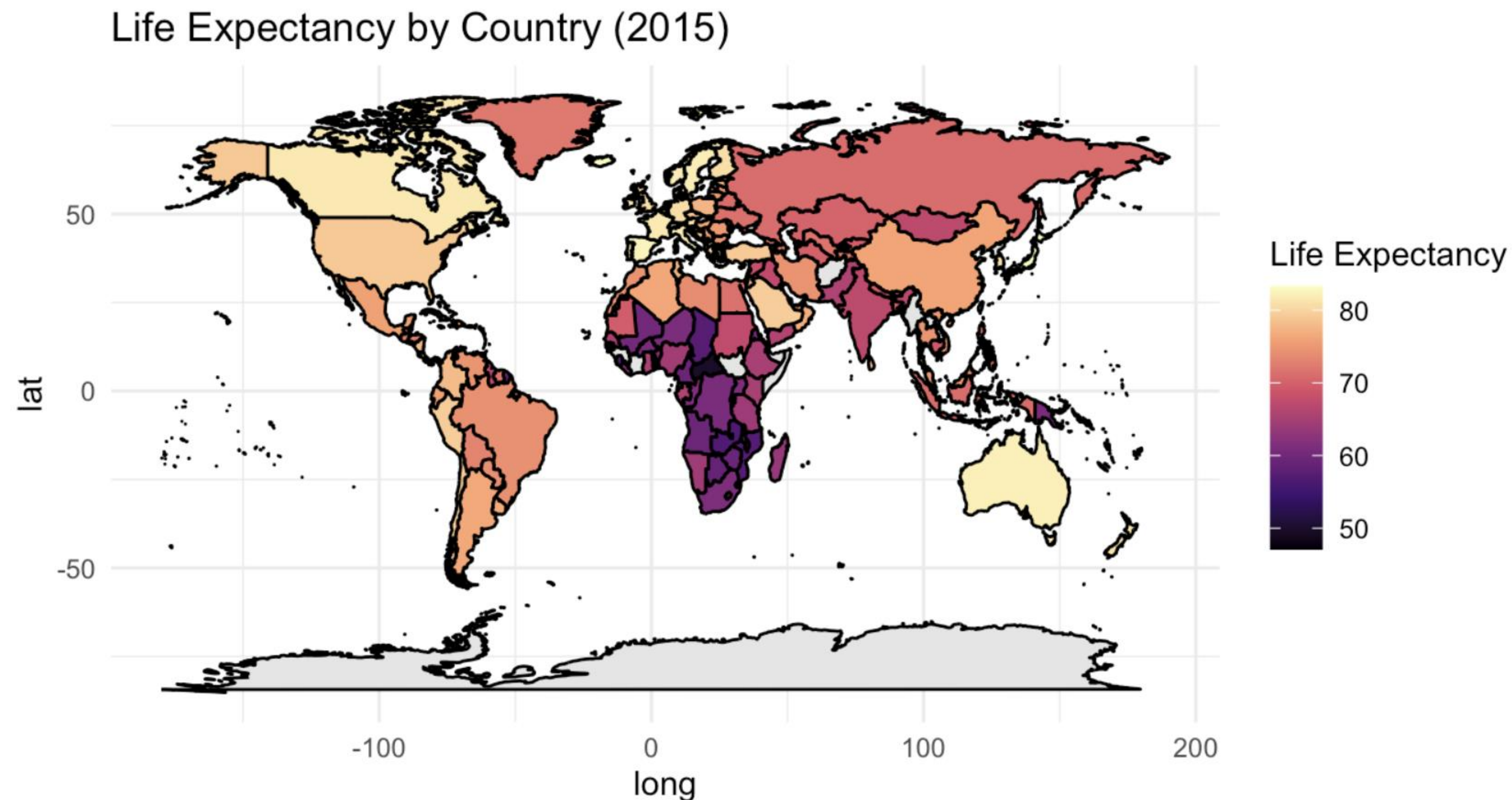


Renaming

```
gapminder_2015 <- gapminder %>% filter(year == 2015) %>%  
  mutate(country =  
    revalue(country, c("Antigua and Barbuda" = "Antigua",  
                        "Samoa" = "American Samoa",  
                        "Congo, Rep." = "Republic of Congo",  
                        "Congo, Dem. Rep." = "Democratic Republic of the Congo",  
                        "Guinea" = "French Guiana",  
                        "Kyrgyz Republic" = "Kyrgyzstan",  
                        "St. Lucia" = "Saint Lucia",  
                        "St. Vincent and the Grenadines" = "Saint Vincent",  
                        "Trinidad and Tobago" = "Trinidad",  
                        "United Kingdom" = "UK",  
                        "United States" = "USA")))
```

```
world_map <- map_data("world")  
merged_world <- left_join(world_map, gapminder_2015, by = c("region" = "country"))
```

```
ggplot(merged_world, aes(long, lat, group = group, fill = life_expectancy)) +  
  geom_polygon(color = "black") +  
  scale_fill_viridis_c(option = "magma", na.value = "gray90") +  
  theme_minimal() +  
  labs(title = "Life Expectancy by Country (2015)", fill = "Life Expectancy") +  
  coord_fixed(1.3)
```



Summary

- Inconsistencies in the spelling of locations can lead to missing data in plots of maps
- Renaming the locations in one data frame before joining to a second data frame can solve a lot of inconsistencies
- Sometimes new data will need to be collected to complete a map plot

Maps

Plotting US state murder rates

Heather Mattie, PhD

Plotting state murder rates

- Task: plot a map of the US states and fill in each state with a color representing the murder rate in that state using the **murders** dataset
- Need to use state map data and add the murder rate for each state to the map data frame

```
library(tidyverse)
library(maps)
library(dslabs)
library(RColorBrewer)

data(murders)
murders <- murders %>% mutate(rate = total / population * 100000)
states_map <- map_data("state")

head(murders)
```

```
##      state abb region population total   rate
## 1  Alabama AL  South   4779736   135 2.824424
## 2  Alaska  AK   West    710231    19 2.675186
## 3  Arizona AZ   West   6392017   232 3.629527
## 4  Arkansas AR  South   2915918    93 3.189390
## 5 California CA   West  37253956  1257 3.374138
## 6  Colorado CO   West   5029196    65 1.292453
```

```
head(states_map)
```

```
##      long      lat group order  region subregion
## 1 -87.46201 30.38968     1     1 alabama    <NA>
## 2 -87.48493 30.37249     1     2 alabama    <NA>
## 3 -87.52503 30.37249     1     3 alabama    <NA>
## 4 -87.53076 30.33239     1     4 alabama    <NA>
## 5 -87.57087 30.32665     1     5 alabama    <NA>
## 6 -87.58806 30.32665     1     6 alabama    <NA>
```


Plotting state murder rates

- Task: plot a map of the US states and fill in each state with a color representing the murder rate in that state using the **murders** dataset
- Need to use state map data and add the murder rate for each state to the map data frame

```
murders <- murders %>% mutate(region = tolower(state))  
head(murders)
```

```
##      state abb  region population total    rate  
## 1  Alabama AL   alabama  4779736  135 2.824424  
## 2   Alaska AK   alaska   710231   19 2.675186  
## 3  Arizona AZ   arizona  6392017  232 3.629527  
## 4 Arkansas AR  arkansas  2915918   93 3.189390  
## 5 California CA california 37253956 1257 3.374138  
## 6  Colorado CO   colorado  5029196   65 1.292453
```

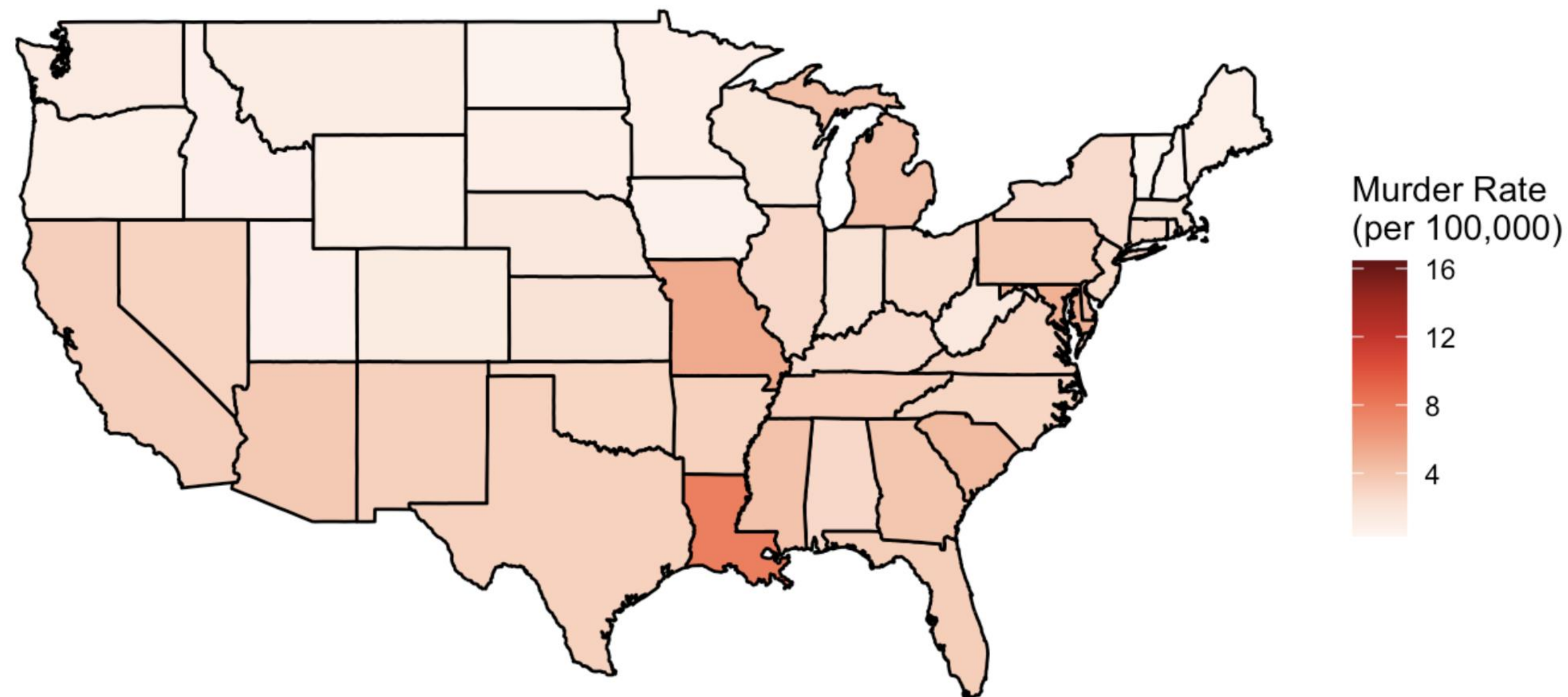
```
merged_data <- left_join(states_map, murders, by = "region")  
head(merged_data)
```

```
##      long    lat group order region subregion state abb population total  
## 1 -87.46201 30.38968    1     1 alabama      <NA> Alabama AL   4779736  135  
## 2 -87.48493 30.37249    1     2 alabama      <NA> Alabama AL   4779736  135  
## 3 -87.52503 30.37249    1     3 alabama      <NA> Alabama AL   4779736  135  
## 4 -87.53076 30.33239    1     4 alabama      <NA> Alabama AL   4779736  135  
## 5 -87.57087 30.32665    1     5 alabama      <NA> Alabama AL   4779736  135  
## 6 -87.58806 30.32665    1     6 alabama      <NA> Alabama AL   4779736  135  
##      rate  
## 1 2.824424  
## 2 2.824424  
## 3 2.824424  
## 4 2.824424  
## 5 2.824424  
## 6 2.824424
```

Plotting state murder rates

```
merged_data %>%  
  ggplot(aes(long, lat, group = group, fill = rate)) +  
    geom_polygon(color = "black") +  
    scale_fill_gradientn(colors = brewer.pal(9, "Reds")) +  
    theme_void() +  
    labs(title = "US Murder Rate by State",  
         fill = "Murder Rate \n(per 100,000)") +  
    coord_fixed(1.3)
```

US Murder Rate by State



Adding state labels

```
library(ggrepel)

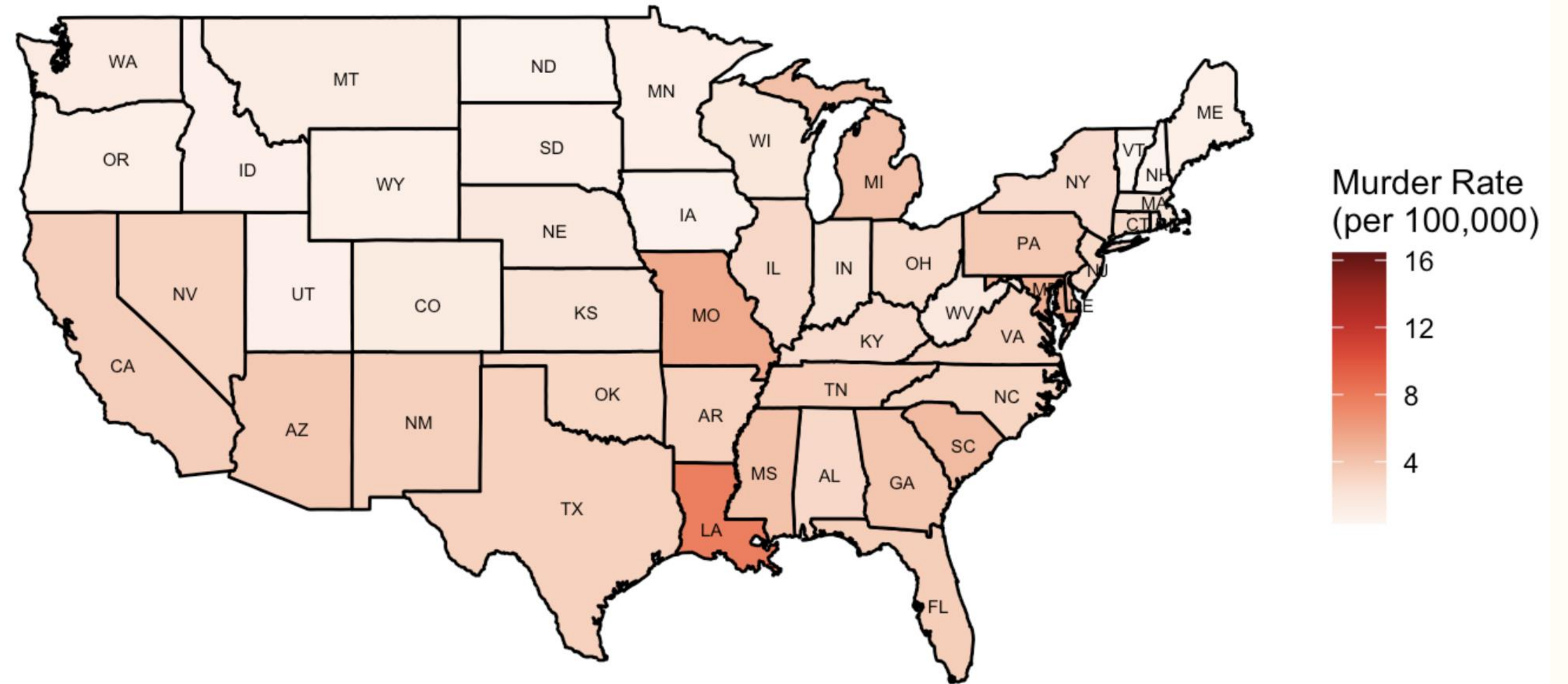
centroids <- data.frame(region = tolower(state.name),
                        long = state.center$x,
                        lat = state.center$y)

centroids$abb <- state.abb[match(centroids$region,
                                tolower(state.name))]

centroids <- centroids %>%
  filter(!region %in% c("alaska", "hawaii"))

merged_data %>%
  ggplot(aes(long, lat, group = group, fill = rate)) +
  geom_polygon(color = "black") +
  scale_fill_gradientn(colors = brewer.pal(9, "Reds")) +
  theme_void() +
  labs(title = "US Murder Rate by State",
       fill = "Murder Rate \n(per 100,000)") +
  coord_fixed(1.3) +
  with(centroids,
       annotate(geom = "text", x = long, y = lat, label = abb,
               size = 2, color = "black", family = "Arial"))
```

US Murder Rate by State



Summary

- `map_data("states")` can be used to create a map of the contiguous United States
- The `tolower` function makes all letters in a string lowercase
- `state.center$x` and `state.center$y` contain the centroid longitude and latitude points for each US state
- Centroid locations can be used to plot state labels

Maps

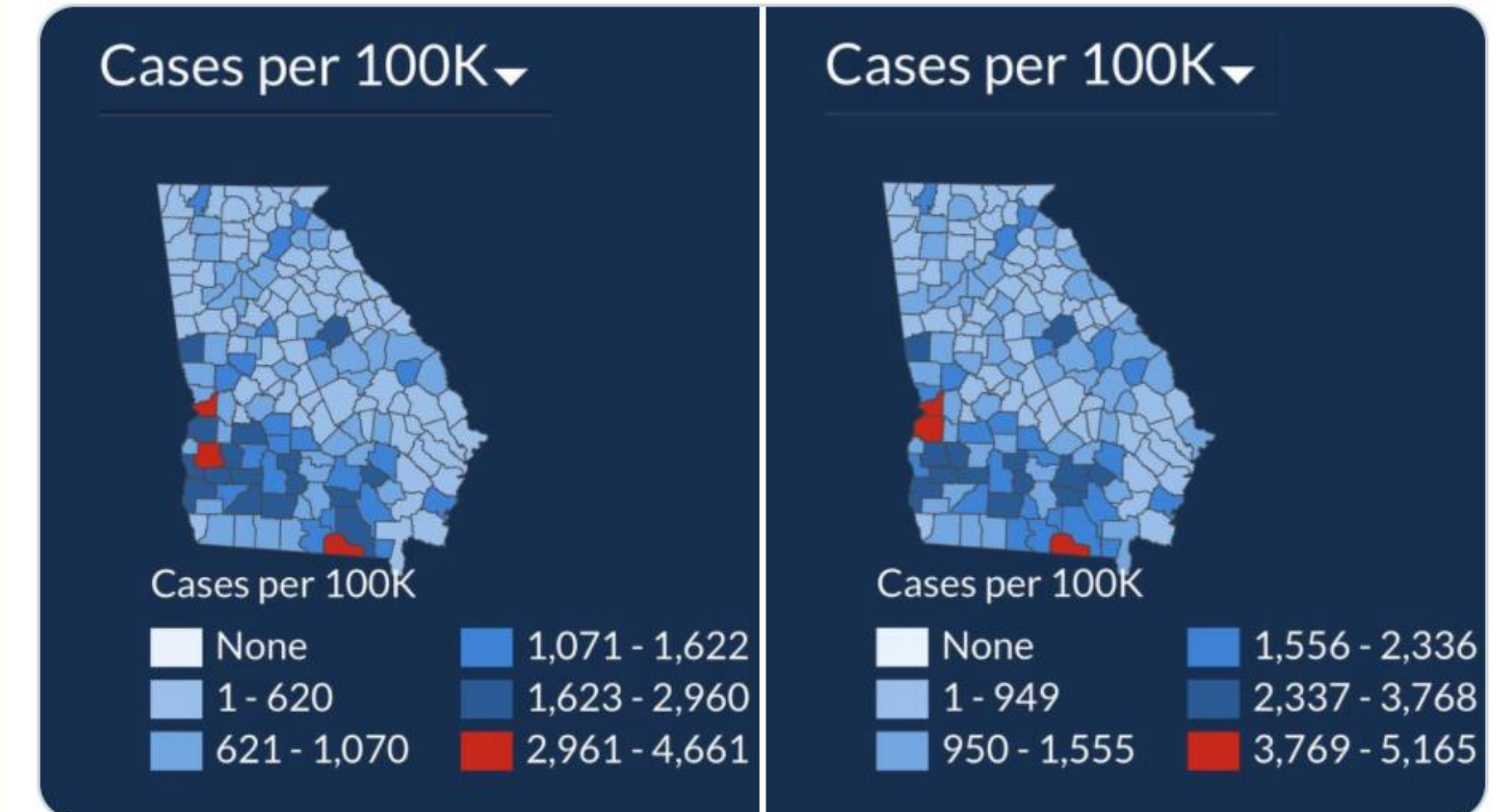
Case study: COVID-19

Heather Mattie, PhD

Misleading visualization

- In 2020 the Georgia Department of Public Health posted daily map visualizations of COVID-19 case counts in each county in Georgia
- On July 17, 2020, a tweet comparing the map on two different days pointed out an issue

In just 15 days the total number of [#COVID19](#) cases in Georgia is up 49%, but you wouldn't know it from looking at the state's data visualization map of cases. The first map is July 2. The second is today. Do you see a 50% case increase? Can you spot how they're hiding it? 1/

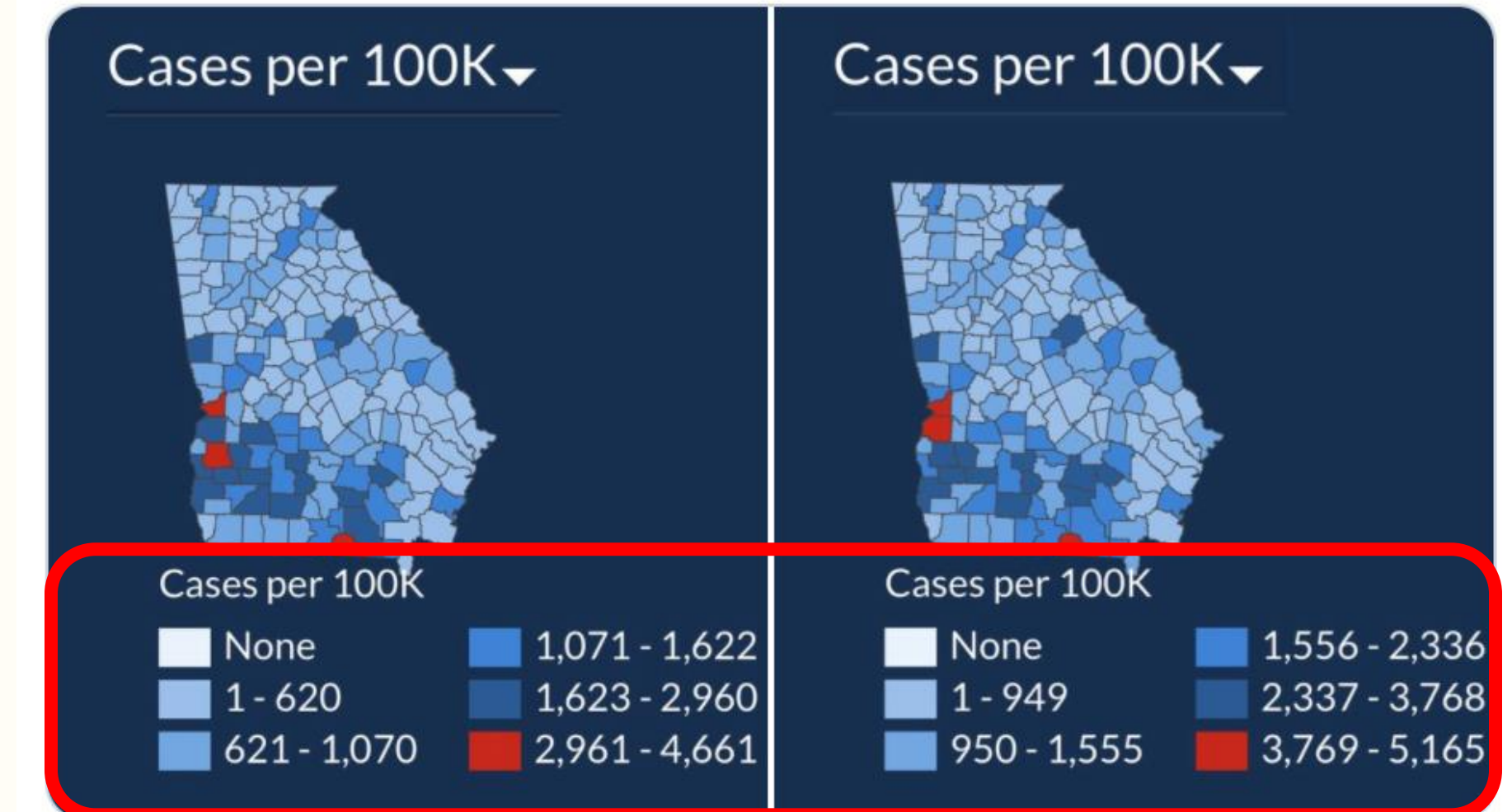


5:24 PM · Jul 17, 2020 · [Twitter for iPhone](#)

Misleading visualization

- In 2020 the Georgia Department of Public Health posted daily map visualizations of COVID-19 case counts in each county in Georgia
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In just 15 days the total number of [#COVID19](#) cases in Georgia is up 49%, but you wouldn't know it from looking at the state's data visualization map of cases. The first map is July 2. The second is today. Do you see a 50% case increase? Can you spot how they're hiding it? 1/



5:24 PM · Jul 17, 2020 · Twitter for iPhone

Data

We will need to combine three different datasets to recreate this plot

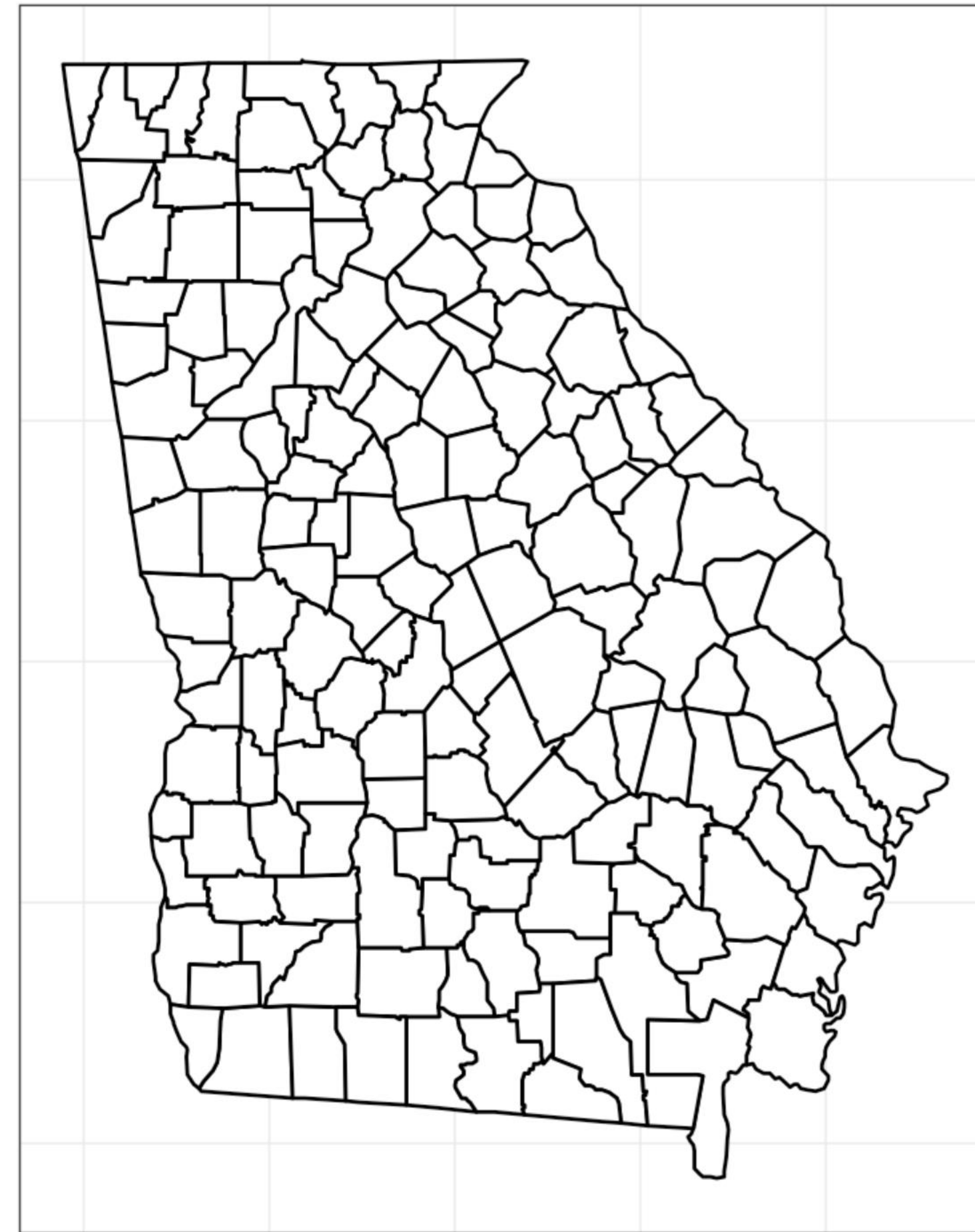
- Georgia county map data
 - From the **maps** package
- Daily COVID-19 case counts for each Georgia county
 - From New York Times COVID-19 data repository
- Population of each Georgia county
 - From the Georgia Department of Public Health (saved as csv)

Georgia county map data

```
GeorgiaCounty <- map_data("county", "georgia")  
head(GeorgiaCounty)
```

```
##           long      lat group order  region subregion  
## 1 -82.44862 31.94813     1      1 georgia  applying  
## 2 -82.42570 31.94813     1      2 georgia  applying  
## 3 -82.40852 31.94240     1      3 georgia  applying  
## 4 -82.39706 31.94240     1      4 georgia  applying  
## 5 -82.38560 31.93094     1      5 georgia  applying  
## 6 -82.35122 31.91948     1      6 georgia  applying
```

```
GeorgiaCounty %>%  
  ggplot(aes(x = long, y = lat, group = group)) +  
  geom_polygon(fill = "white", color = "black") +  
  theme(panel.grid.major = element_blank(),  
        panel.background = element_blank(),  
        axis.title = element_blank(),  
        axis.text = element_blank(),  
        axis.ticks = element_blank()) +  
  coord_fixed(1.3)
```



COVID-19 data

nytimes / covid-19-data

Q Type / to search

+

<> Code

Issues 2

Pull requests

Projects

Security

Insights

covid-19-data

Public

Watch 317

Fork 3.4k

Star 7k

master

2 Branches

0 Tags

Go to file

Add file

<> Code

nyt-covid-19-bot

Updating data.

62ef34c · 2 years ago

4,101 Commits

.github/ISSUE_TEMPLATE	New data for 6/26.	5 years ago
colleges	update colleges readme to note it is not currently being u...	4 years ago
excess-deaths	Final update for excess deaths dataset.	5 years ago
live	Updating data.	2 years ago
mask-use	*NEW DATASET*: Estimates of mask-usage by county fro...	5 years ago
prisons	NEW: We're publishing files detailing the outbreak in priso...	4 years ago
rolling-averages	Updating data.	2 years ago
.gitignore	excess deaths	5 years ago
LICENSE	Update license and citation year	5 years ago

About

A repository of data on coronavirus cases and deaths in the U.S.

[www.nytimes.com/interactive/2020/u...](#)

covid-19

Readme

View license

Activity

Custom properties

7k stars

317 watching

3.4k forks

Report repository

Releases



COVID-19 data

```
url <- "https://raw.githubusercontent.com/nytimes/covid-19-data/master/us-counties.csv"
cases <- read_csv(url)
head(cases)
```

```
## # A tibble: 6 × 6
##   date      county    state    fips  cases deaths
##   <date>    <chr>    <chr>    <chr> <dbl>  <dbl>
## 1 2020-01-21 Snohomish Washington 53061     1      0
## 2 2020-01-22 Snohomish Washington 53061     1      0
## 3 2020-01-23 Snohomish Washington 53061     1      0
## 4 2020-01-24 Cook      Illinois   17031     1      0
## 5 2020-01-24 Snohomish Washington 53061     1      0
## 6 2020-01-25 Orange    California 06059     1      0
```

COVID-19 data

```
dates <- c(ymd("2020-07-02", "2020-07-17"))
georgia_cases <- cases %>% filter(state == "Georgia", date %in% dates)
head(georgia_cases)
```

```
## # A tibble: 6 × 6
##   date      county  state  fips  cases  deaths
##   <date>    <chr>   <chr> <chr> <dbl>  <dbl>
## 1 2020-07-02 Appling  Georgia 13001   266    14
## 2 2020-07-02 Atkinson Georgia 13003   155     2
## 3 2020-07-02 Bacon    Georgia 13005   254     4
## 4 2020-07-02 Baker    Georgia 13007    43     3
## 5 2020-07-02 Baldwin Georgia 13009   543    34
## 6 2020-07-02 Banks    Georgia 13011   140     1
```


Georgia county population data

```
pop_data <- read_csv("county_pop.csv")
head(pop_data)
```

```
## # A tibble: 6 × 12
##   county_name cases county_id `State FIPS code` `County FIPS code` population
##   <chr>      <dbl> <chr>      <chr>          <chr>          <dbl>
## 1 Appling      1054 US-13001  13              1              18561
## 2 Atkinson      447 US-13003  13              3               8330
## 3 Bacon         602 US-13005  13              5              11404
## 4 Baker          84 US-13007  13              7               3116
## 5 Baldwin     2132 US-13009  13              9              44428
## 6 Banks         501 US-13011  13             11              19982
## # i 6 more variables: hospitalization <dbl>, deaths <dbl>, `case rate` <dbl>,
## #   `death rate` <dbl>, `14 day case rate` <dbl>, `14 day cases` <dbl>
```

Joining data frames

1. Join the county population data to the COVID-19 cases data

```
pop_data <- pop_data %>% select(county_name, population)
```

```
data_full <- left_join(georgia_cases, pop_data, by = c("county" = "county_name"))  
head(data_full)
```

```
## # A tibble: 6 × 7  
##   date      county  state  fips  cases  deaths  population  
##   <date>    <chr>   <chr> <chr> <dbl>  <dbl>    <dbl>  
## 1 2020-07-02 Appling  Georgia 13001   266     14    18561  
## 2 2020-07-02 Atkinson Georgia 13003   155      2     8330  
## 3 2020-07-02 Bacon    Georgia 13005   254      4    11404  
## 4 2020-07-02 Baker    Georgia 13007    43      3     3116  
## 5 2020-07-02 Baldwin  Georgia 13009   543     34    44428  
## 6 2020-07-02 Banks    Georgia 13011   140      1    19982
```

Joining data frames

- 1. Join the county population data to the COVID-19 cases data
- 2. Calculate cases per 100,000 population

```
data_full <- data_full %>% mutate(rate = 100000*(cases/population))
head(data_full)
```

```
## # A tibble: 6 × 8
```

	date	county	state	fips	cases	deaths	population	rate
	<date>	<chr>	<chr>	<chr>	<dbl>	<dbl>	<dbl>	<dbl>
## 1	2020-07-02	Appling	Georgia	13001	266	14	18561	1433.
## 2	2020-07-02	Atkinson	Georgia	13003	155	2	8330	1861.
## 3	2020-07-02	Bacon	Georgia	13005	254	4	11404	2227.
## 4	2020-07-02	Baker	Georgia	13007	43	3	3116	1380.
## 5	2020-07-02	Baldwin	Georgia	13009	543	34	44428	1222.
## 6	2020-07-02	Banks	Georgia	13011	140	1	19982	701.



Joining data frames

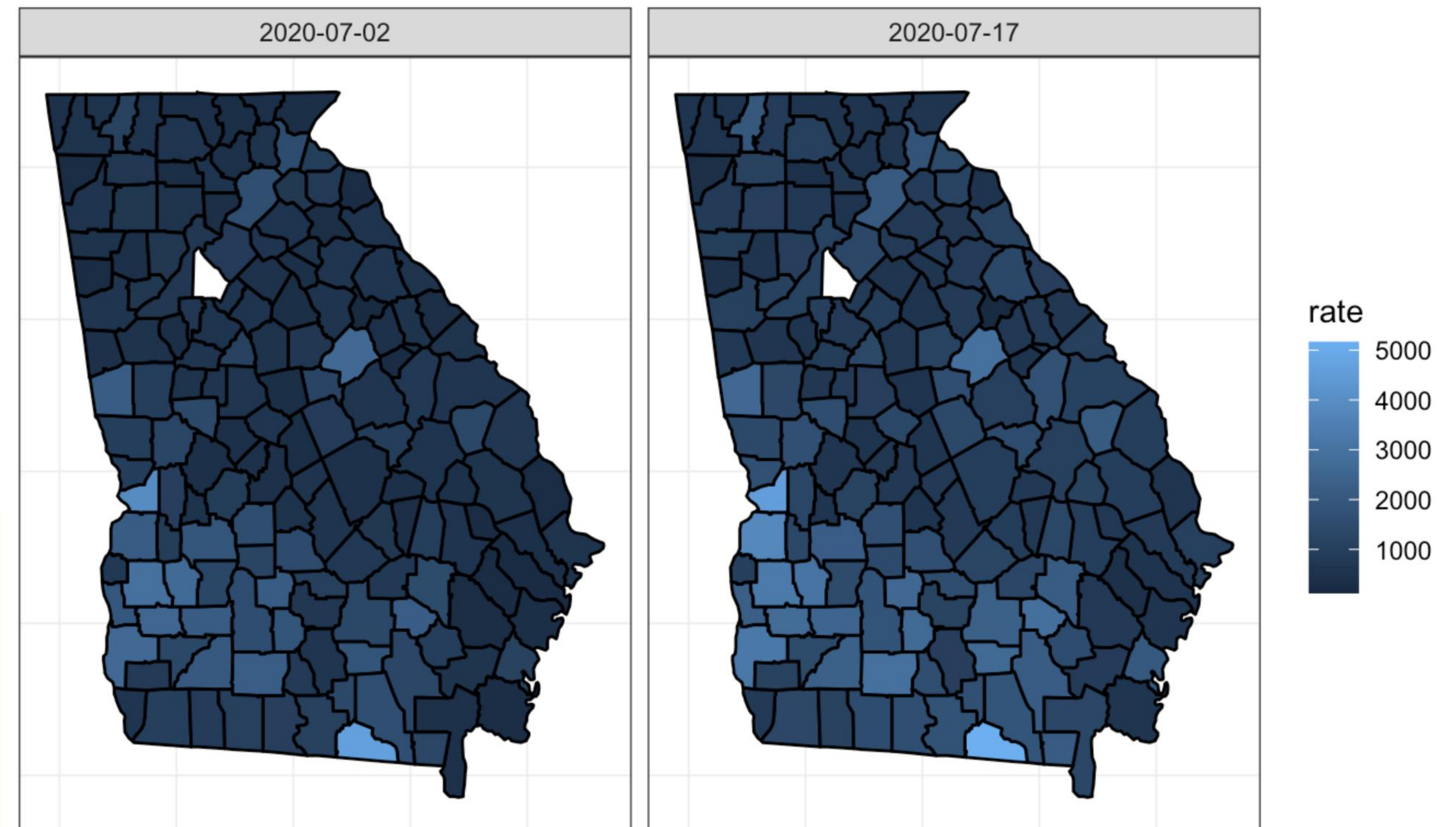
- 1. Join the county population data to the COVID-19 cases data
- 2. Calculate cases per 100,000 population
- 3. Join to map data

```
georgia_map <- data_full %>% mutate(county = str_to_lower(county)) %>%  
  left_join(GeorgiaCounty, by = c("county" = "subregion"))  
  
head(georgia_map)
```

```
## # A tibble: 6 × 13  
##   date      county state fips  cases deaths population  rate  long  lat group  
##   <date>    <chr>  <chr> <chr> <dbl>  <dbl>      <dbl> <dbl> <dbl> <dbl> <dbl>  
## 1 2020-07-02 appling Geor... 13001   266    14      18561 1433. -82.4  31.9    1  
## 2 2020-07-02 appling Geor... 13001   266    14      18561 1433. -82.4  31.9    1  
## 3 2020-07-02 appling Geor... 13001   266    14      18561 1433. -82.4  31.9    1  
## 4 2020-07-02 appling Geor... 13001   266    14      18561 1433. -82.4  31.9    1  
## 5 2020-07-02 appling Geor... 13001   266    14      18561 1433. -82.4  31.9    1  
## 6 2020-07-02 appling Geor... 13001   266    14      18561 1433. -82.4  31.9    1  
## # i 2 more variables: order <int>, region <chr>
```

Data inconsistencies

```
georgia_map %>%  
  ggplot(aes(x = long, y = lat, group = group, fill = rate)) +  
  geom_polygon(color = "black") +  
  theme(panel.grid.major = element_blank(),  
        panel.background = element_blank(),  
        axis.title = element_blank(),  
        axis.text = element_blank(),  
        axis.ticks = element_blank()) +  
  coord_fixed(1.3) +  
  facet_grid( ~ date)
```

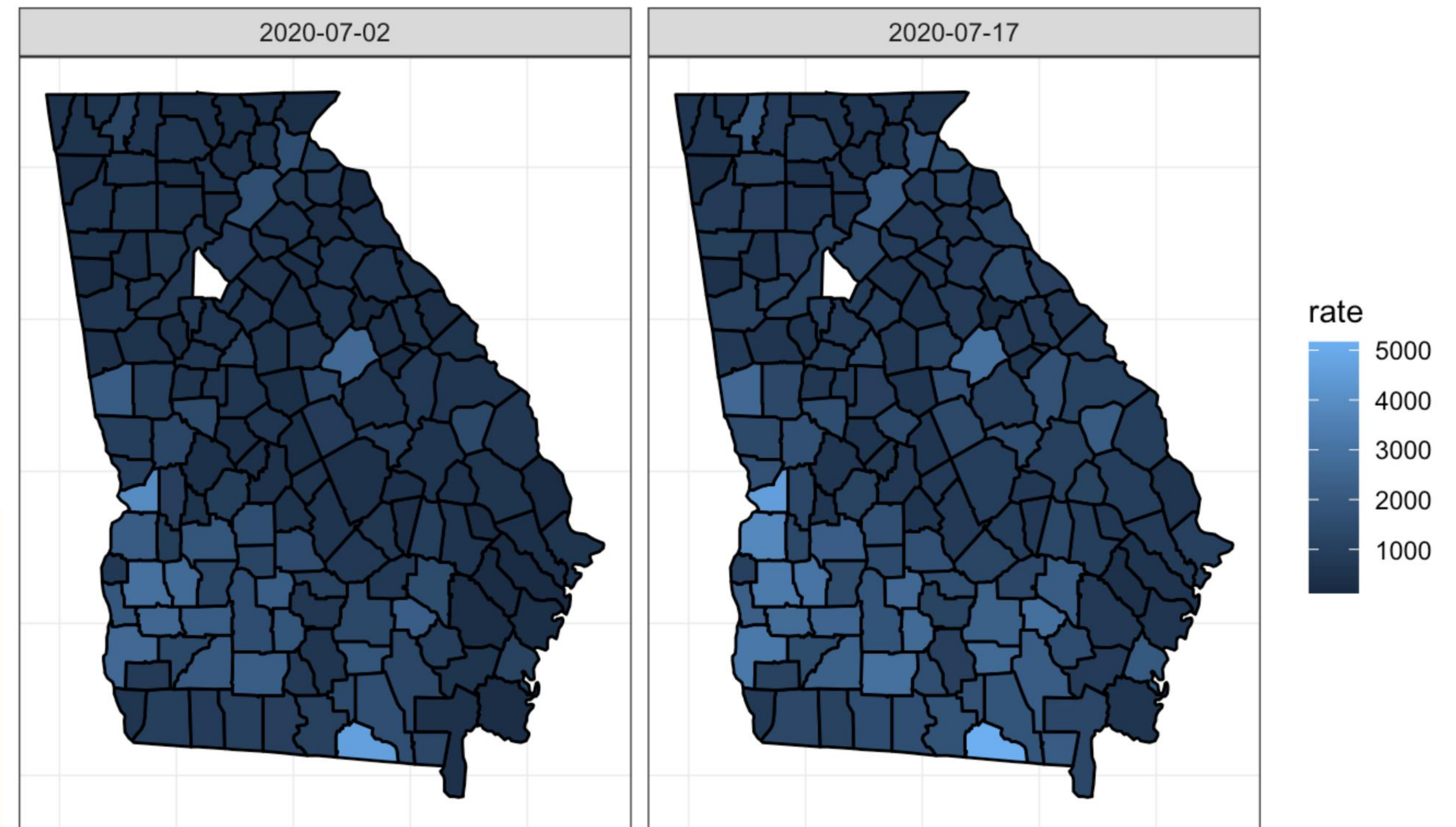


Data inconsistencies

```
georgia_map %>%  
  ggplot(aes(x = long, y = lat, group = group, fill = rate)) +  
  geom_polygon(color = "black") +  
  theme(panel.grid.major = element_blank(),  
        panel.background = element_blank(),  
        axis.title = element_blank(),  
        axis.text = element_blank(),  
        axis.ticks = element_blank()) +  
  coord_fixed(1.3) +  
  facet_grid( ~ date)
```

Problem: De Kalb county

Saved as “**de kalb**” in
the **GeorgiaCounty** data frame, but as
“**DeKalb**” in the in the **georgia_cases**
data frame



Data inconsistencies

First solution attempt:

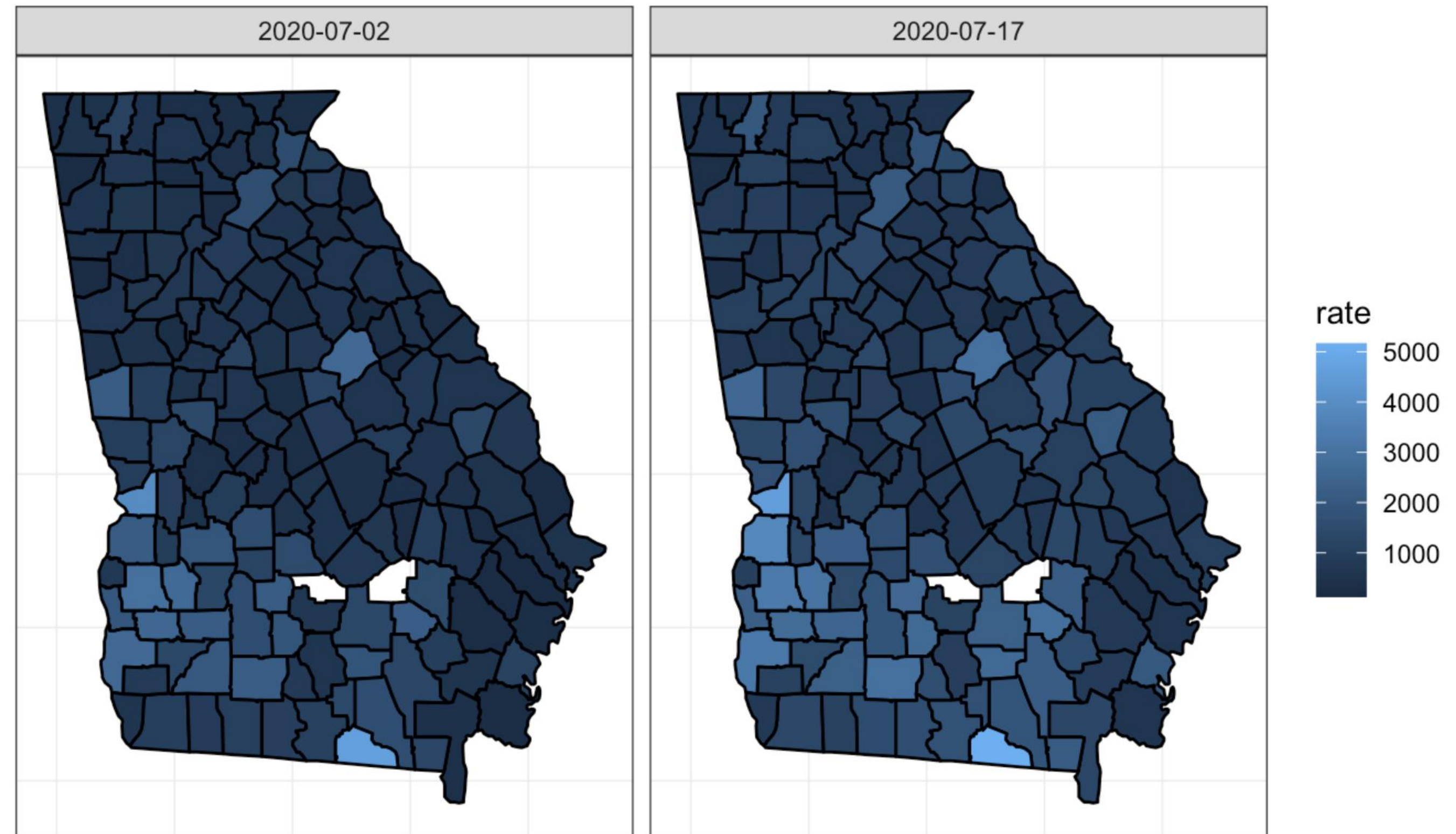
1. Remove all spaces in county names using the **str_replace** function
2. Convert all county names to lowercase using the **str_to_lower** function

```
GeorgiaCounty <- GeorgiaCounty %>%  
  mutate(subregion = str_replace(subregion, " ", ""))  
  
georgia_map <- data_full %>%  
  mutate(county = str_to_lower(county)) %>%  
  left_join(GeorgiaCounty, by = c("county" = "subregion"))  
  
head(georgia_map)
```

```
## # A tibble: 6 × 13  
##   date      county state fips  cases deaths population  rate  long  lat group  
##   <date>    <chr>  <chr> <chr> <dbl>  <dbl>      <dbl> <dbl> <dbl> <dbl> <dbl>  
## 1 2020-07-02 applying Geor... 13001    266     14      18561 1433. -82.4  31.9     1  
## 2 2020-07-02 applying Geor... 13001    266     14      18561 1433. -82.4  31.9     1  
## 3 2020-07-02 applying Geor... 13001    266     14      18561 1433. -82.4  31.9     1  
## 4 2020-07-02 applying Geor... 13001    266     14      18561 1433. -82.4  31.9     1  
## 5 2020-07-02 applying Geor... 13001    266     14      18561 1433. -82.4  31.9     1  
## 6 2020-07-02 applying Geor... 13001    266     14      18561 1433. -82.4  31.9     1  
## # i 2 more variables: order <int>, region <chr>
```

Data inconsistencies

```
georgia_map %>%  
  ggplot(aes(x = long, y = lat, group = group, fill = rate)) +  
  geom_polygon(color = "black") +  
  theme(panel.grid.major = element_blank(),  
        panel.background = element_blank(),  
        axis.title = element_blank(),  
        axis.text = element_blank(),  
        axis.ticks = element_blank()) +  
  coord_fixed(1.3) +  
  facet_grid( ~ date)
```

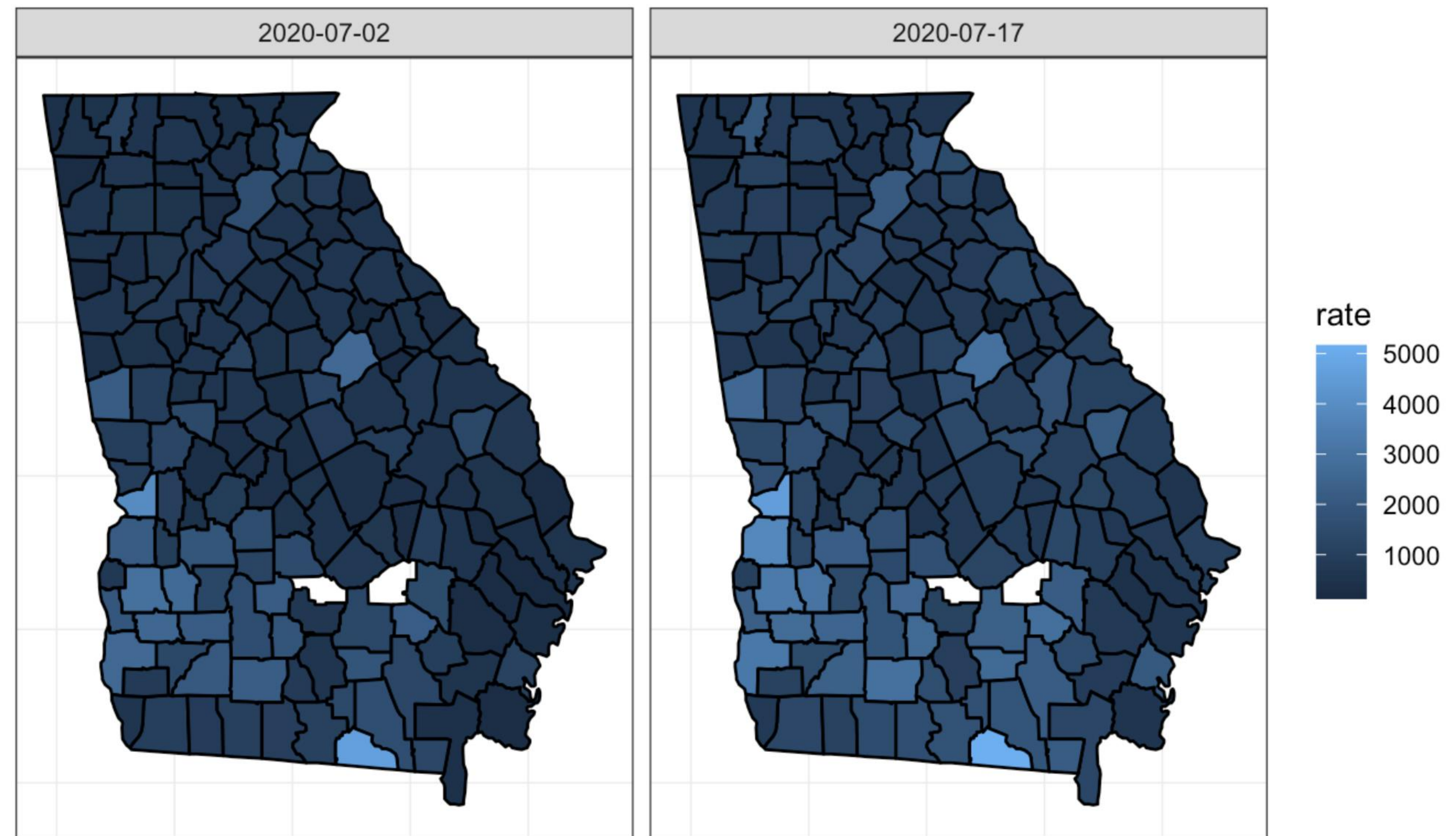


Data inconsistencies

```
georgia_map %>%  
  ggplot(aes(x = long, y = lat, group = group, fill = rate)) +  
  geom_polygon(color = "black") +  
  theme(panel.grid.major = element_blank(),  
        panel.background = element_blank(),  
        axis.title = element_blank(),  
        axis.text = element_blank(),  
        axis.ticks = element_blank()) +  
  coord_fixed(1.3) +  
  facet_grid( ~ date)
```

Problem: **Ben Hill** and **Jeff Davis** counties

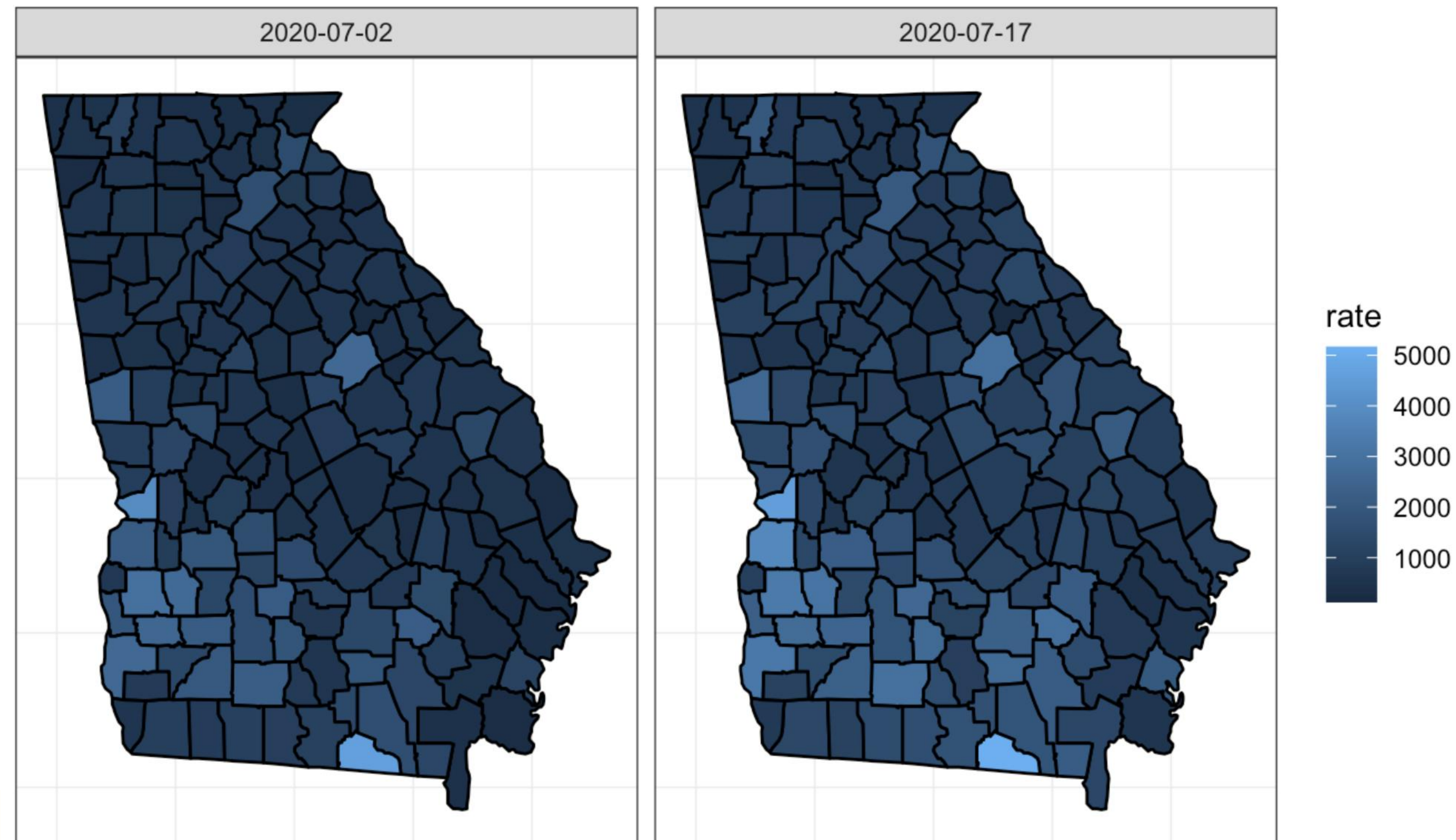
Both data frames had a space between the two names but the space was removed for **GeorgiaCounty** and not for **georgia_cases**



Data inconsistencies

```
data_full <- data_full %>%  
  mutate(county = str_replace(county, " ", ""))  
  
georgia_map <- data_full %>%  
  mutate(county = str_to_lower(county)) %>%  
  left_join(GeorgiaCounty, by = c("county" = "subregion"))
```

```
georgia_map %>%  
  ggplot(aes(x = long, y = lat, group = group, fill = rate)) +  
  geom_polygon(color = "black") +  
  theme(panel.grid.major = element_blank(),  
        panel.background = element_blank(),  
        axis.title = element_blank(),  
        axis.text = element_blank(),  
        axis.ticks = element_blank()) +  
  coord_fixed(1.3) +  
  facet_grid( ~ date)
```



Customizing legends

```
georgia_map <- georgia_map %>%  
  mutate(manual_fill = cut(rate, breaks = c(0, 620, 1070, 1622, 2960, Inf),  
    labels = c("1-620", "621-1,070", "1,071-1,622",  
      "1,623-2,960", ">2,960"),  
    right = TRUE))  
  
head(georgia_map)
```

```
## # A tibble: 6 × 14  
##   date      county state fips  cases deaths population  rate  long  lat group  
##   <date>    <chr>  <chr> <chr> <dbl>  <dbl>      <dbl> <dbl> <dbl> <dbl> <dbl>  
## 1 2020-07-02 appling Geor... 13001   266    14      18561 1433. -82.4  31.9     1  
## 2 2020-07-02 appling Geor... 13001   266    14      18561 1433. -82.4  31.9     1  
## 3 2020-07-02 appling Geor... 13001   266    14      18561 1433. -82.4  31.9     1  
## 4 2020-07-02 appling Geor... 13001   266    14      18561 1433. -82.4  31.9     1  
## 5 2020-07-02 appling Geor... 13001   266    14      18561 1433. -82.4  31.9     1  
## 6 2020-07-02 appling Geor... 13001   266    14      18561 1433. -82.4  31.9     1  
## # i 3 more variables: order <int>, region <chr>, manual_fill <fct>
```

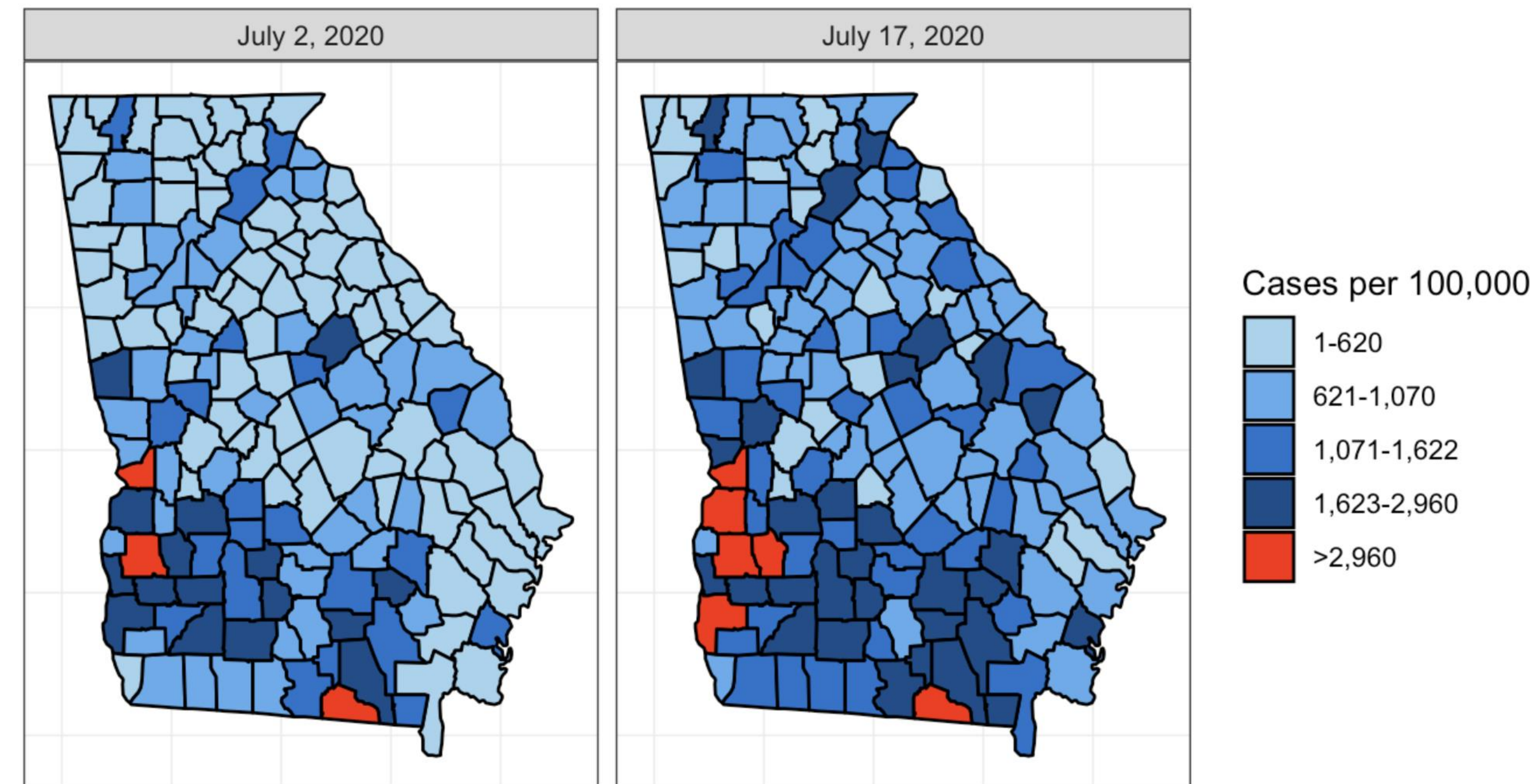

Customizing color palettes

```
pal <- c("lightskyblue2", "steelblue2", "dodgerblue3", "dodgerblue4", "red")
```

```
date.labs <- c("July 2, 2020", "July 17, 2020")  
names(date.labs) <- c("2020-07-02", "2020-07-17")
```

```
georgia_map %>%  
  ggplot(aes(x = long, y = lat, group = group)) +  
  geom_polygon(aes(fill = manual_fill), color = "black") +  
  scale_fill_manual(name = "Cases per 100,000", values = pal) +  
  coord_fixed(1.3) +  
  theme(panel.grid.major = element_blank(),  
        panel.background = element_blank(),  
        axis.title = element_blank(),  
        axis.text = element_blank(),  
        axis.ticks = element_blank()) +  
  ggtitle("COVID-19 Cases per 100K") +  
  facet_grid(. ~ date, labeller = labeller(date = date.labs))
```

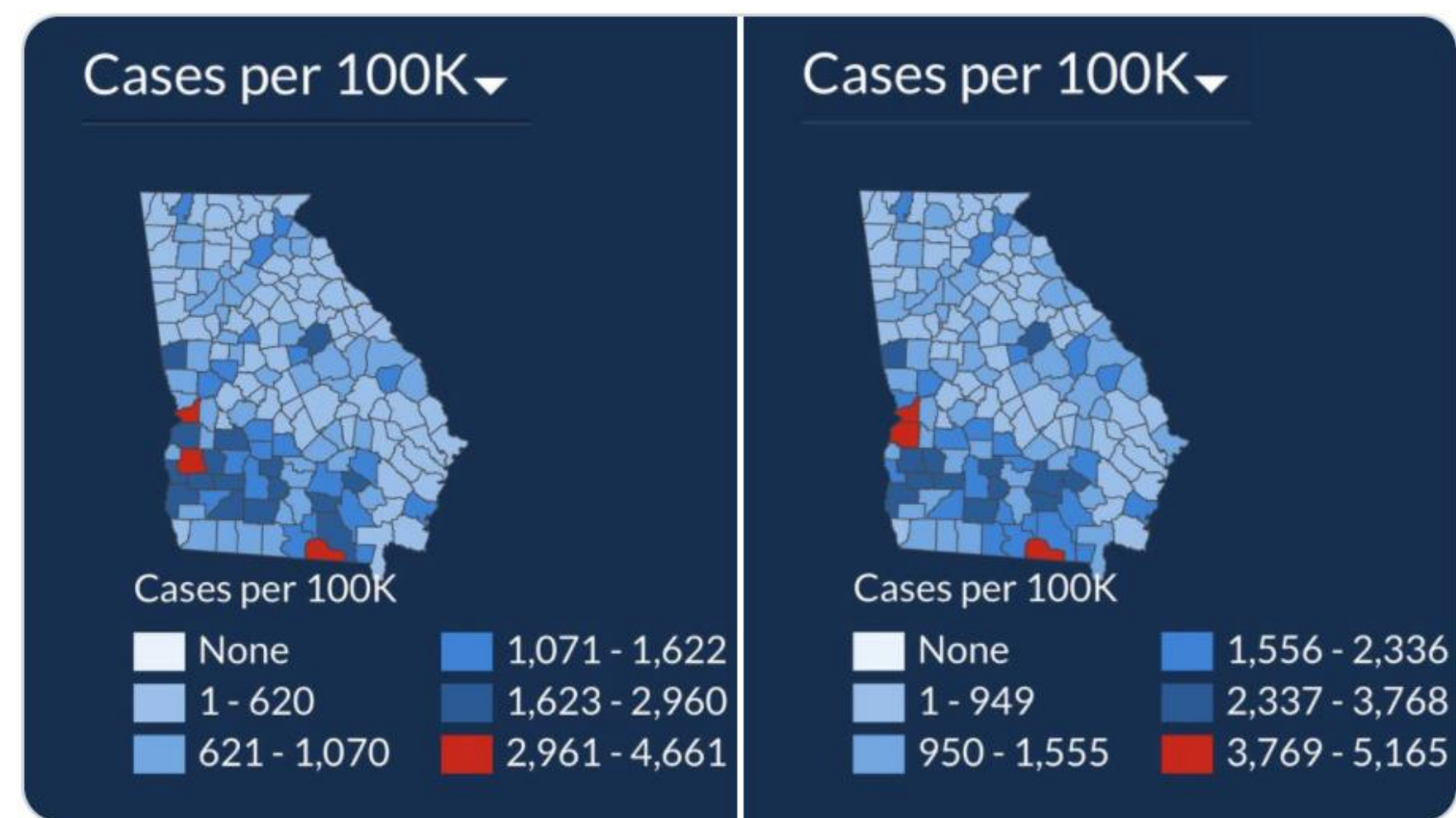
COVID-19 Cases per 100K



Plot comparison

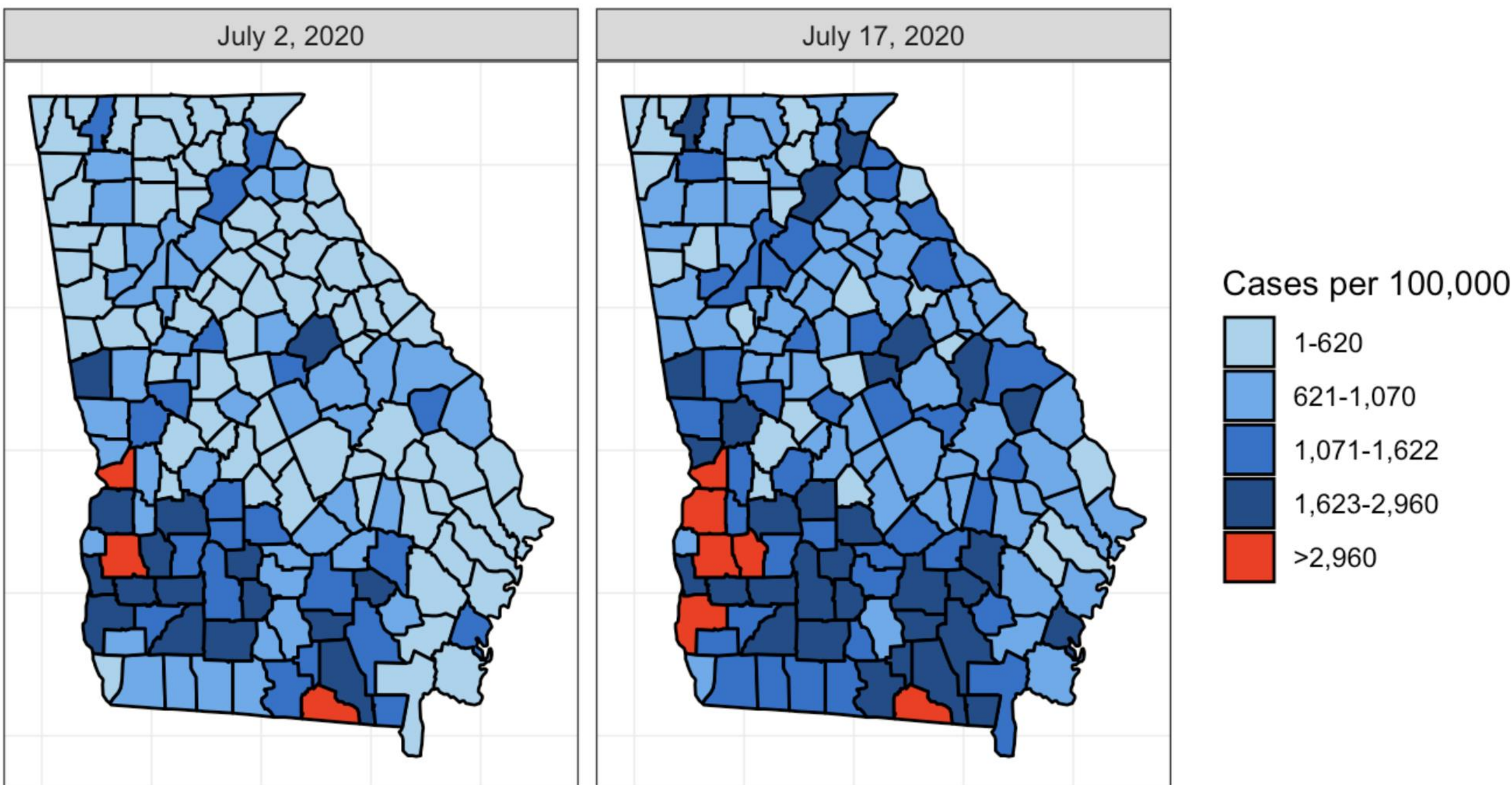


In just 15 days the total number of [#COVID19](#) cases in Georgia is up 49%, but you wouldn't know it from looking at the state's data visualization map of cases. The first map is July 2. The second is today. Do you see a 50% case increase? Can you spot how they're hiding it? 1/



5:24 PM · Jul 17, 2020 · [Twitter for iPhone](#)

COVID-19 Cases per 100K



Summary

- Plots of maps can be misleading, especially when plotting changes over time
- There should only be one legend per plot if subplots are related
- Color palettes can be customized